PROJECT REPORT

BMS Project 1: Predictive Modelling of mAb Fucosylation in CHO Cell Bioprocessing

A comparative study of machine learning models on synthetic, mechanistically-grounded bioprocess data with a fully automated end-to-end pipeline.

Industry Partner:

Bristol-Myers Squibb (BMS)

Authors

Sanjith Ganesh (sg2151)

Pranav Senthilkumaran (ps1471)

M.S. Data Science · Rutgers University

May 2026

Abstract

Fucosylation is one of the most consequential post-translational modifications a monoclonal antibody (mAb) can carry. Whether or not a fucose sugar sits on the core of the antibody's Fc N-glycan changes how strongly it binds to Fcγ receptors on immune cells, and that single bond can shift antibody-dependent cellular cytotoxicity (ADCC) by an order of magnitude. For therapeutic antibodies produced in Chinese hamster ovary (CHO) cells — which is essentially the entire commercial mAb market — fucosylation is therefore a critical quality attribute (CQA) tracked closely under ICH Q8/Q9 guidelines. The bioprocessing variables that drive it (substrate availability, cofactors, pH, dissolved oxygen, temperature) are all controllable in a bioreactor, but the relationships between them are nonlinear, interacting, and noisy.

This project asks a deceptively simple question: given a stack of standard bioprocess measurements, how well can different machine learning models predict the resulting fucosylation level — and which model is right for the job? We benchmark seven models spanning the interpretability–accuracy spectrum (Ridge, PLSR, Random Forest, ANN, GPR, a Hybrid physics-informed model, and XGBoost+SHAP) on two synthetic datasets that were generated from a mechanistic ground truth grounded in the published biochemistry literature. The two datasets — referred to throughout this report as BMS1 (N = 500) and BMS2 (N = 10,000) — differ only in scale, allowing us to isolate the effect of dataset size on model behaviour.

The headline findings are clean. Linear models hit a structural ceiling near $R^2 = 0.57$ regardless of data volume — they cannot capture the quadratic pH and temperature optima or the GDP-Fucose × Mn^{2+} synergy. Tree ensembles (Random Forest, XGBoost) are robust at both data scales; XGBoost+SHAP achieves the best overall performance at N = 10,000 ($R^2 = 0.82$) while remaining fully explainable. Gaussian Process Regression dominates the small-data regime ($R^2 = 0.75$ at N = 500) with calibrated uncertainty, but its $O(n^3)$ cost forces a 300-sample subsample at N = 10,000. The ANN tells the most dramatic story: it fails at N = 500 ($R^2 \approx 0.05$, badly overparameterised) and recovers to $R^2 = 0.81$ at N = 10,000 — a 1,440% improvement that quantifies the 'data hunger' of unconstrained neural networks. The Hybrid model demonstrates that five physics-informed engineered features alone can capture roughly three-quarters of full model performance, validating domain knowledge as a dimensionality-reduction tool.

The work described here is the second of three planned stages. Stage 1 — an automated dataset cleansing module that ingests any BMS-supplied dataset and normalises it to the schema expected by the model pipeline. Stage 3 — a stakeholder-facing dashboard that exposes model results, feature attributions, and uncertainty bounds in a non-technical format — is also in progress and likewise documented separately. This report focuses on Stage 2: the model bench, the synthetic data design that supports it, and the comparative results.

Contents

1. Introduction & Motivation

Therapeutic monoclonal antibodies are now a roughly 200-billion-dollar global industry, and the overwhelming majority of them are manufactured in Chinese hamster ovary (CHO) cell lines. CHO cells are popular for a reason — they are easy to engineer, they grow well in suspension, and they produce human-compatible glycosylation patterns. That last point is more important than it sounds. The sugar chains attached to an antibody's Fc region are not cosmetic; they are functional. They control how long the molecule stays in circulation, how it engages immune cells, and ultimately how potent the drug is in a patient.

Fucosylation — specifically core α-1,6 fucosylation catalysed by the enzyme FUT8 — is the glycan modification with the largest published effect on mechanism of action. Antibodies missing core fucose (afucosylated) bind FcγRIIIa with up to ~50× higher affinity, producing dramatically stronger ADCC. This is exactly why molecules like obinutuzumab (Gazyva) and mogamulizumab were engineered to be afucosylated, and it is why fucosylation level is monitored on every batch in commercial mAb manufacturing as part of the Quality-by-Design (QbD) framework laid out in ICH Q8.

Predicting fucosylation from upstream bioprocess parameters is therefore a high-value problem. If a model can tell a process engineer 'this combination of pH, temperature, dissolved oxygen, and feed strategy will land you at 92% fucosylation', the engineer can adjust before the batch goes wrong. The catch is that the relationships are nonlinear (pH and temperature both have well-defined optima with quadratic penalties on either side), interacting (GDP-Fucose substrate concentration and Mn^{2+} cofactor synergise multiplicatively at the FUT8 active site), and obscured by batch effects and measurement noise. Linear models leave a lot of variance on the table; black-box models leave regulators uncomfortable. The right answer almost always involves comparing several modelling families and choosing one that fits both the data and the regulatory context.

That comparison is the core of this project. The work was scoped as a three-stage pipeline:

- **Stage 1 — Automated dataset cleansing.** A module that ingests any dataset BMS provides and normalises it (missing-value handling, outlier filtering, schema enforcement, feature scaling) into the format expected by the model bench. This stage is in progress and is being written up by Pranav Senthilkumaran.
- **Stage 2 — Comparative model bench.** The seven-model benchmark that this report documents. Two datasets, identical mechanistic structure, only the sample size and batch count differ. Every model runs through the same train/test split, the same scaler, and the same evaluation rubric.
- **Stage 3 — Stakeholder dashboard.** A visual layer (Power BI / web dashboard) that surfaces the model bench's outputs — performance comparisons, SHAP feature attributions, uncertainty bounds, learning curves — in a form a non-technical stakeholder can read in a meeting. This stage is also in progress and is being written up by Pranav Senthilkumaran.

The narrative of this report is therefore: introduce the dataset (Section 3), describe the seven models that act on it (Section 4), report each model's outputs on the small dataset and then the large dataset (Sections 5 and 6), pull the cross-model comparison together (Section 7), and place the model bench in the broader pipeline context (Sections 8–9, written up by Pranav).

2. Dataset Design

2.1 Why Synthetic Data — and Why It Is Defensible

Real CHO bioprocessing data is proprietary. BMS does not publish it, and the closest open analogues (academic CHO datasets) tend to be small, narrow in process scope, and missing the glycan readouts. Rather than silently hand-wave this away, we built a synthetic dataset where every variable, every coefficient, and every functional form is grounded in published biochemistry. Three references do most of the heavy lifting:

- **Raju et al. (2000)** — establishes GDP-fucose as the direct substrate for the FUT8 fucosyltransferase. Justifies making GDP_Fucose_uM the strongest single predictor and giving it Michaelis–Menten saturation rather than a linear effect.
- **Hossler et al. (2009)** — shows that media components such as manganese, dissolved oxygen, and osmolality measurably change CHO glycosylation patterns. Justifies the inclusion of Mn_uM, DO_pct, Osmolality_mOsm, and Lactate_g_per_L as predictors and the directional sign on each.
- **Jedrzejewski et al. (2014)** — builds mechanistic models of CHO metabolism that link culture conditions to product quality. Justifies the overall structural choice of starting from a mechanistic equation and adding noise on top, rather than starting from random correlations.

Two additional sources — Sha et al. (2016) on Golgi processing and Von Stosch et al. (2014) on hybrid mechanistic-data models — informed the engineered-feature design used in the Hybrid model (Section 4.7).

The synthetic dataset is therefore best understood as a test bench for the modelling pipeline. The advantage of synthetic data is that the ground truth is known, so we can verify whether each model recovers the right relationships rather than just chasing test-set R^2. When the pipeline is later run on real BMS data, that ground-truth scaffolding is gone — but the pipeline itself has been validated on a problem where the true coefficients can be checked against what the models report.

2.2 Input Variables and Biological Roles

The dataset has ten input variables, each chosen because it is both biologically meaningful and routinely measured in a CHO bioreactor. The ranges below are taken from Hossler et al. (2009) and from typical fed-batch operating envelopes.

Variable	Range / Units	Biological role
GDP_Fucose_uM	5 – 80 µM	Direct substrate for FUT8 (primary driver, saturating)
Mn_uM	0.001 – 0.05 µM	Glycosyltransferase cofactor — synergises with GDP-Fucose
Uridine_mM	0.5 – 5.0 mM	Nucleotide-sugar precursor (mild positive)
pH	6.6 – 7.4	Quadratic optimum at 7.0; enzyme denatures at extremes
DO_pct	20 – 80 %	Dissolved oxygen — supports oxidative metabolism
pCO2_mmHg	30 – 120 mmHg	Disrupts intracellular pH at high levels (negative)

Variable	Range / Units	Biological role
Temperature_C	33 – 38 °C	Quadratic optimum at 36.5 °C
VCD_1e6_per_mL	2 – 25	Viable cell density — weak positive
Osmolality_mOsm	280 – 400 mOsm/kg	High osmolality → cellular stress (mild negative)
Lactate_g_per_L	0.1 – 4.0 g/L	Metabolic-byproduct stress marker (negative)

Table 1. *Input variables, plausible operating ranges, and biological role.*

2.3 Ground-Truth Mechanistic Equation

The 'true' fucosylation percentage for a sample is computed from the input variables using the equation below. Inputs are first min–max normalised to [0, 1] so the coefficients have direct interpretive meaning (each coefficient is the percentage-point change in fucosylation as that variable moves from its minimum to its maximum).

$$
\begin{aligned}
\text{Fuc\%} = 85.0 \\
&+ 12.0 \cdot \left(\widetilde{GDP} / (\widetilde{GDP} + 0.3) \right) && \text{\# Michaelis–Menten saturation} \\
&+ 4.0 \cdot \widetilde{Mn} \quad + 2.0 \cdot \widetilde{Uridine} && \text{\# cofactor + precursor} \\
&- 8.0 \cdot ((pH - 7.0)/0.4)^2 && \text{\# quadratic pH optimum} \\
&- 5.0 \cdot ((T - 36.5)/2.5)^2 && \text{\# quadratic temperature optimum} \\
&+ 3.0 \cdot \widetilde{DO} \quad - 4.0 \cdot \widetilde{pCO_2} && \text{\# gas exchange} \\
&+ 1.5 \cdot \widetilde{VCD} \quad - 2.5 \cdot \widetilde{Osm} \quad - 3.0 \cdot \widetilde{Lac} && \text{\# cells / stress / waste} \\
&+ 3.5 \cdot \widetilde{GDP} \cdot \widetilde{Mn} && \text{\# cofactor synergy}
\end{aligned}
$$

Tilde notation indicates the min–max normalised version of each variable (mean of zero is not assumed; the [0, 1] interval is the relevant biological span). The intercept of 85% is calibrated so a sample at average operating conditions lands in the typical commercial fucosylation range. The two quadratic terms (pH, temperature) and the GDP-Fucose × Mn^{2+} interaction term are the explicit nonlinear structure that linear models will fail to capture.

2.4 Batch Effects, Heteroscedastic Noise, and Missingness

A clean ground truth alone is not realistic enough to test models against. Three layers of perturbation are added on top:

- **Batch effects.** Each sample is assigned a batch ID, and each batch gets a random intercept drawn from N(0, 1.5) — i.e. a ±1.5% shift specific to that batch. This is a stand-in for equipment, operator, and lot-to-lot variation that always shows up in industrial bioprocessing data.
- **Heteroscedastic measurement noise.** The base measurement noise is N(0, 1.8) on the percentage scale, with an additive heteroscedastic term that scales with how far the sample is from the dataset mean. This is faithful to assay behaviour at extreme glycan compositions.
- **Missingness — MCAR and MNAR.** Two missing-data variants are generated. The MCAR variant drops 5% of feature values uniformly. The MNAR variant simulates realistic sensor failure: DO probes failing 30% of the time when DO > 70%, and pCO_2 probes drifting 25% of

the time when pCO_2 > 100 mmHg. The MNAR set is the more useful test of an imputation strategy, because the pattern of missingness itself carries information.

In addition to the primary file, four noise-corrupted variants are saved (low / medium / high / extreme) at noise multipliers of 0.5×, 1.0×, 2.0×, and 4.0× the base level. These are used by Script 08 to map each model's robustness curve.

2.5 Time-Series Variant (Culture Day Progression)

A separate time-series file (mab_fucosylation_timeseries.csv) is generated to capture intra-batch dynamics over a 14-day fed-batch run. Within each batch the substrates deplete (GDP-Fucose decays exponentially), pCO_2 accumulates, pH drifts down as lactate builds, and viable cell density follows a logistic growth curve. This file is not used directly by Stage 2's seven-model bench (which is steady-state) but it is part of the dataset deliverable so that downstream time-series modelling can plug straight in.

2.6 BMS1 vs BMS2 — The Single-Variable Experiment

The two datasets are deliberately constructed so that only sample size differs. The mechanistic equation, the noise process, the missing-data process, and every coefficient are identical. The only changes are:

Parameter	BMS1 (small)	BMS2 (large)
N_SAMPLES (primary)	500	10,000
N_BATCHES (random intercepts)	10	50
N_BATCHES_TS (time-series file)	20	100
Everything else	identical	identical

Table 2. *The only three configuration differences between the BMS1 and BMS2 dataset generators.*

This is a deliberate experimental design choice. By holding the data-generating process fixed and varying only the sample count, any difference we see between BMS1 and BMS2 model performance can be attributed cleanly to data volume — not to a richer signal, not to a different noise structure, not to better feature engineering. It is the simplest possible answer to the question 'do my models actually need more data, or are they at the structural ceiling?'

2.7 Alignment with Bristol-Myers Squibb Practice

Several design decisions were made specifically to keep the dataset (and therefore the model outputs) consistent with how an industrial sponsor like Bristol-Myers Squibb would actually use the pipeline:

- **Variables that map to BMS process analytical technology (PAT) tags.** Every input variable in this dataset is something that is either logged automatically by a typical bioreactor control system (pH, DO, pCO_2, temperature, VCD, osmolality, lactate) or measured offline through standard QC assays (GDP-Fucose, Mn, uridine). Nothing here requires custom instrumentation.
- **ICH Q8/Q9-aligned ranges.** The operating ranges in Table 1 fall inside the typical Design Space described in ICH Q8 for fed-batch CHO processes, which is the regulatory framework BMS files under.

- **Batch effects, not just i.i.d. noise.** Industrial datasets are clustered by batch — that is the dominant source of structured variability — so the dataset includes a random-intercept batch term, which is also how mixed-effects models in pharma typically handle this.
- **Realistic sensor-failure missingness (MNAR).** The MNAR pattern (DO probe failure at high DO, pCO_2 drift at high pCO_2) was chosen because both are well-documented failure modes in pharma bioreactor instrumentation. It is the kind of missingness pattern Stage 1's cleansing module needs to handle.
- **Interpretability-first model selection.** The rubric in Section 7 explicitly scores each model on regulatory fitness (ICH Q8/Q9/Q10 alignment) and uncertainty quantification, not just R^2. This reflects what BMS's QbD and PAT teams care about.

4. Models & Methods (Common to BMS1 and BMS2)

All seven models in this study are trained, evaluated, and reported using the same protocol. The only thing that changes between BMS1 and BMS2 is the dataset. The model code itself — including hyperparameters — is identical between the two folders. (One small, scale-driven exception is noted in Section 4.6: the GPR's training subsample cap is held at 300 in both runs, because GPR is $O(n^3)$ and would otherwise blow up at N = 10,000.)

4.1 Train/Test Protocol

Every model uses the same 80/20 split of the primary dataset (mab_fucosylation_dataset.csv), with a fixed random seed of 42. Features are standardised with sklearn's StandardScaler fit on the training set only. The target variable is Fucosylation_pct, kept on its native percentage scale. The four metrics reported throughout are R^2_train, R^2_test, RMSE_test, and MAE_test, with 5-fold cross-validation R^2 added where it makes sense.

4.2 Model 1 — Ridge Regression (Linear Baseline)

Ridge is the linear baseline. It is what every other model should be measured against — if a complex model cannot beat Ridge, the complexity is not buying anything. The implementation is sklearn's RidgeCV with α searched over fifty values logarithmically spaced from 10^{-4} to 10^{4}, picked by 5-fold cross-validation. OLS is also fit alongside for comparison; the two come out essentially identical because the problem is well-conditioned (all VIFs < 1.05).

Ridge is fully interpretable — each standardised coefficient directly states how much the predicted fucosylation moves per one standard deviation of that input. Diagnostic plots check residual normality (Shapiro–Wilk), homoscedasticity (Breusch–Pagan), VIF for multicollinearity, and Cook's distance for influential observations.

4.3 Model 2 — PLSR (Partial Least Squares Regression)

PLSR is the chemometrics gold standard for spectroscopic and bioprocess data. Where Ridge regularises by shrinking, PLSR projects the input space onto a small set of latent components that maximally covary with the target, which both regularises and gives a built-in feature-importance ranking via VIP scores. The number of components is selected by 5-fold CV, sweeping from 1 to 10. PLSR is explicitly named in pharma PAT/QbD guidance, which is why it scores so well on the regulatory axis in Section 7.

4.4 Model 3 — Random Forest

Random Forest is the first model in the bench that can actually capture nonlinear effects — quadratic optima, saturation, threshold behaviour, and interactions all fall out of tree splitting without needing to be specified. The RF is configured with 300 estimators, max depth 15, min samples leaf 5, and OOB scoring on. Two flavours of feature importance are reported: MDI (mean decrease in impurity, biased toward high-cardinality features per Strobl et al. 2007) and permutation importance (the unbiased reference). The two are cross-checked via Spearman rank correlation.

4.5 Model 4 — ANN (MLPRegressor with LIME)

The ANN is a 128-64-32 fully-connected network with ReLU activations, Adam optimiser, adaptive learning rate, L2 regularisation ($\alpha = 1e\text{-}4$), early stopping on a 15% validation hold-out, and a patience of 50 iterations with up to 2,000 epochs. Total parameter count is roughly 12,000. LIME is run on a held-out batch of test points to give local explanations, since the network itself is otherwise opaque.

This model is included specifically to demonstrate the data-hunger of unconstrained neural networks. With ~12,000 parameters and only 400 training samples in BMS1, it is massively overparameterised and the early-stopping criterion fires before it can fit. With 8,000 training samples in BMS2 the parameter-to-sample ratio is comfortably below 1:0.5 and the model converges. The contrast is the headline result of the entire study.

4.6 Model 5 — GPR (Gaussian Process Regression)

GPR is the only model in the bench that returns calibrated uncertainty bounds for free, which makes it valuable for ICH Q8 Design Space work where you genuinely want to know how confident a prediction is. The kernel is an RBF (squared exponential) with a WhiteKernel noise term, both length-scale and noise level optimised via three restarts. Three alternative kernels (Matérn 5/2, Matérn 3/2, Rational Quadratic) are also fit and compared in kernel_comparison.csv to show that RBF is not just an arbitrary default.

Important implementation note: GPR has $O(n^3)$ training cost, which becomes prohibitive at $N = 10{,}000$. To keep the comparison fair, the GPR in both BMS1 and BMS2 trains on a fixed 300-sample subsample of the training set (set as MAX_GPR_SAMPLES = 300 in the BMS2 comparison script). Inference is on the full test set. This is also propagated into the data-efficiency curve in Script 08, where the GPR's training subset is capped at 300 even when more data is available — otherwise the noise-sensitivity sweep would never finish.

4.7 Model 6 — Hybrid (Physics-Informed Feature Engineering)

The Hybrid model is the explicit test of whether domain knowledge is worth its weight in features. Five engineered features are built from the ten raw inputs, each grounded in a specific biochemical mechanism:

Feature	Components	Mechanism	Reference
FPI	GDP-Fucose × Mn^{2+}	FUT8 substrate × cofactor activation	Raju 2000; Sha 2016
Energy_Charge	DO, Lactate, pCO_2	Oxidative vs glycolytic balance	Jedrzejewski 2014
Golgi_Proxy	Uridine, pH, Temperature	Nucleotide-sugar supply × Golgi optima	Sha 2016
Stress_Index	Osmolality, Lactate, pCO_2	Synergistic cellular stress	Hossler 2009
Enzyme_Activity	Mn^{2+}, pH, Temperature	Pseudo-Arrhenius × pH bell curve	Raju 2000

Table 3. *The five physics-informed engineered features in the Hybrid model.*

Three Random Forests are then trained: one on the ten raw features only, one on the five engineered features only, and one on the combined 15-feature set. The interesting comparison is engineered-only vs raw-only: it answers the question 'how much performance can I keep with half the variables, if I encode mechanism instead of letting the model rediscover it?'

4.8 Model 7 — XGBoost + SHAP

XGBoost is the workhorse production model. It captures nonlinearities and interactions, scales to large data, and (since Lundberg & Lee 2017) has theoretically-grounded local explanations through TreeSHAP. The configuration is 300 estimators, max depth 5, learning rate 0.1, subsample 0.8 — a fairly conservative tune that resists overfitting on small data while still capturing the structure.

Three SHAP analyses are run: a global feature importance ranking based on mean absolute SHAP value, a SHAP summary plot showing the direction and magnitude of each feature's effect, and a feature-interaction matrix that surfaces the strongest pairwise interactions. The interaction matrix is the analysis where we expect to see the GDP-Fucose × Mn^{2+} synergy that was baked into the ground truth — and we do.

5. Per-Model Results — BMS1 (N = 500)

This section reports each model's outputs on the small dataset (500 samples, 10 batches). For every model we present the headline metrics, the most informative diagnostic plot from results/0X_<model>/, and a paragraph of interpretation. Where applicable the most important supporting CSV (coefficients, VIP scores, SHAP rankings) is also tabulated.

5.1 Ridge Regression

Metric	OLS	Ridge
R^2 (train)	0.634	0.634
R^2 (test)	0.510	0.513
RMSE (test)	3.448	3.437

Metric	OLS	Ridge
MAE (test)	2.855	2.846
Optimal α	—	3.352

Table 4. *Ridge / OLS metrics on BMS1.*

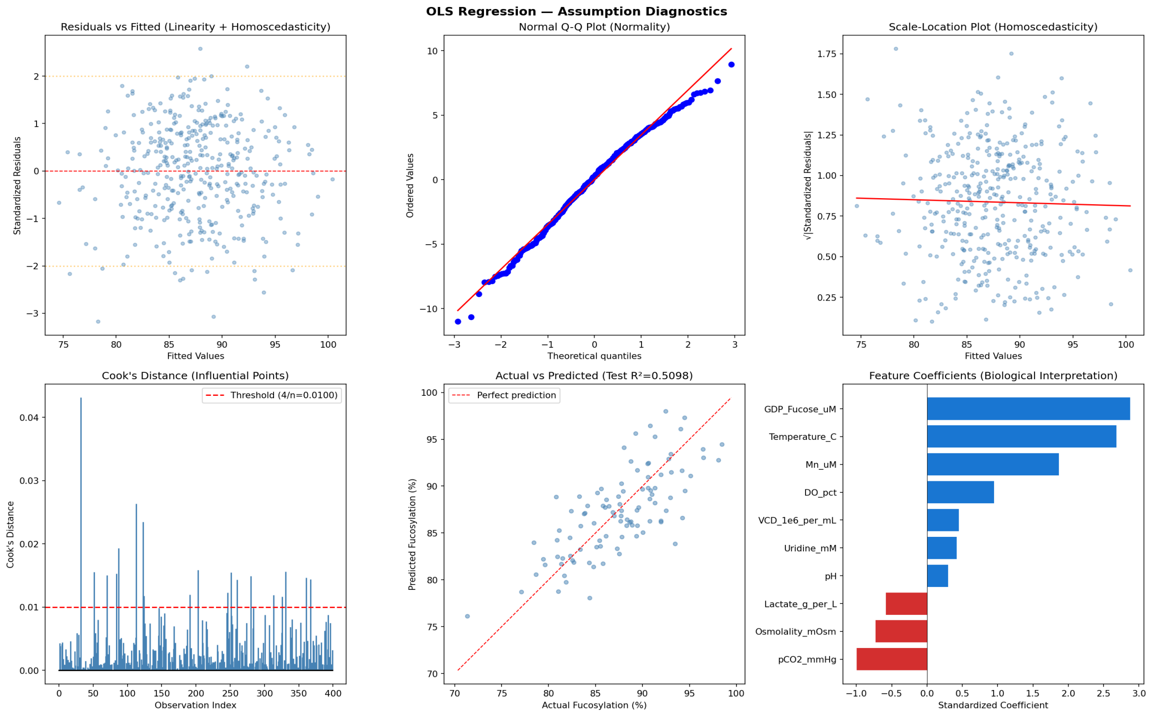


Figure 1. *Ridge regression diagnostic suite on BMS1: residuals vs fitted, Q-Q normality, residual histogram, and predicted vs actual.*

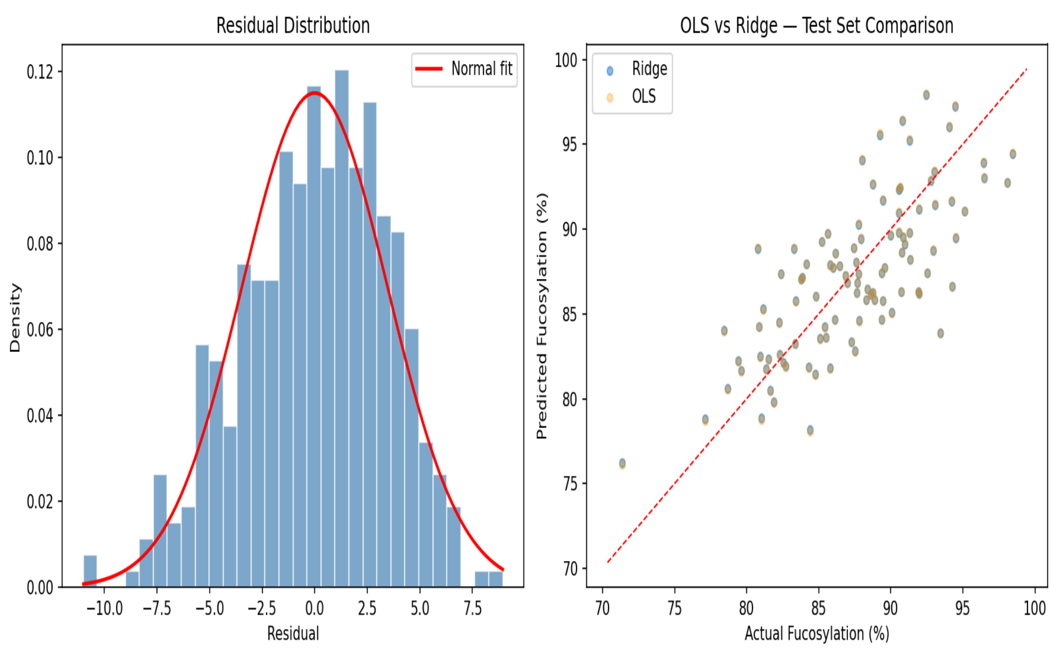


Figure 2. OLS vs Ridge coefficient comparison and α-selection cross-validation curve on BMS1.

Ridge and OLS land on essentially the same answer (test $R^2 = 0.51$), which is exactly what we expect because the problem is well-conditioned (all VIFs sit below 1.05, so multicollinearity is not the bottleneck). The standardised coefficients recover the broad direction of the ground truth — GDP-Fucose and Temperature are the two strongest positive drivers, pCO_2 and Osmolality are negative, and pH is essentially flat (because the linear model averages over the quadratic optimum and gets near-zero net slope). The Breusch–Pagan test shows no heteroscedasticity ($p = 0.44$), Shapiro–Wilk shows mild non-normality ($p = 0.007$ — expected at $n = 500$), and 19 observations cross the Cook's-distance threshold (4.8% of the data) without any single point dominating.

The takeaway: Ridge is a useful interpretive baseline that confirms the linear directional structure, but it leaves substantial variance unexplained. The 49% of variance it cannot capture is exactly the nonlinear and interaction structure the later models target.

5.2 PLSR (Partial Least Squares Regression)

Metric	Value
R^2 (train)	0.634
R^2 (test)	0.510
RMSE (test)	3.447
MAE (test)	2.855
Optimal components	4

Table 5. PLSR metrics on BMS1.

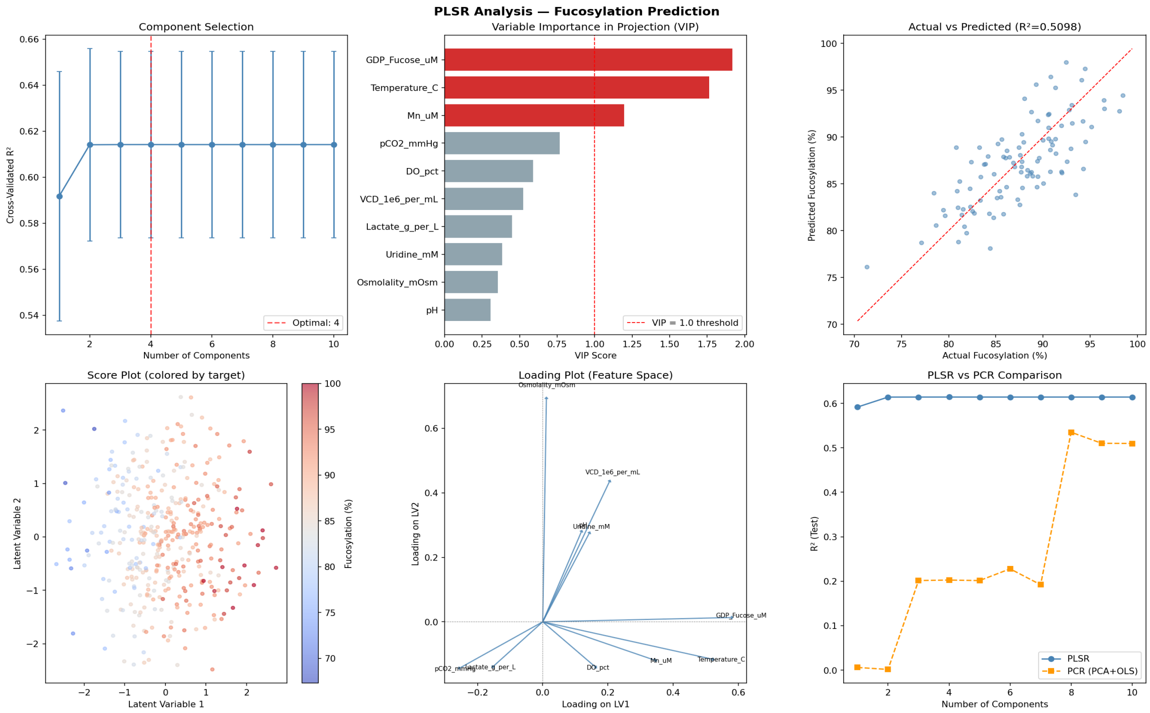


__Figure 3.__ PLSR analysis suite on BMS1: VIP scores, component-selection CV curve, X-loadings, and PLSR vs PCR comparison.

Feature	VIP score
GDP_Fucose_uM	1.917
Temperature_C	1.764
Mn_uM	1.198
pCO2_mmHg	0.767
DO_pct	0.591
VCD_1e6_per_mL	0.523
Lactate_g_per_L	0.449
Uridine_mM	0.384
Osmolality_mOsm	0.355
pH	0.306

__Table 6.__ VIP feature-importance scores from PLSR on BMS1. VIP > 1 indicates a significant contributor.

Cross-validation picks 4 components as optimal — beyond that, additional components stop improving CV-R^2. PLSR converges to the same predictive performance as Ridge ($R^2 = 0.51$), which is consistent with both being members of the linear family. Where PLSR adds value is the VIP scores: three features (GDP-Fucose, Temperature, Mn^{2+}) cross the VIP > 1 threshold and pH falls all the way to 0.306, because the linear projection fails to detect the quadratic pH effect at all. This is a clean, regulator-friendly statement of which inputs matter, and it would be the natural starting point for a PAT submission.

5.3 Random Forest

Metric	Value
R^2 (train)	0.957
R^2 (test)	0.659
RMSE (test)	2.876
MAE (test)	2.198
OOB score	0.716

Table 7. *Random Forest metrics on BMS1.*

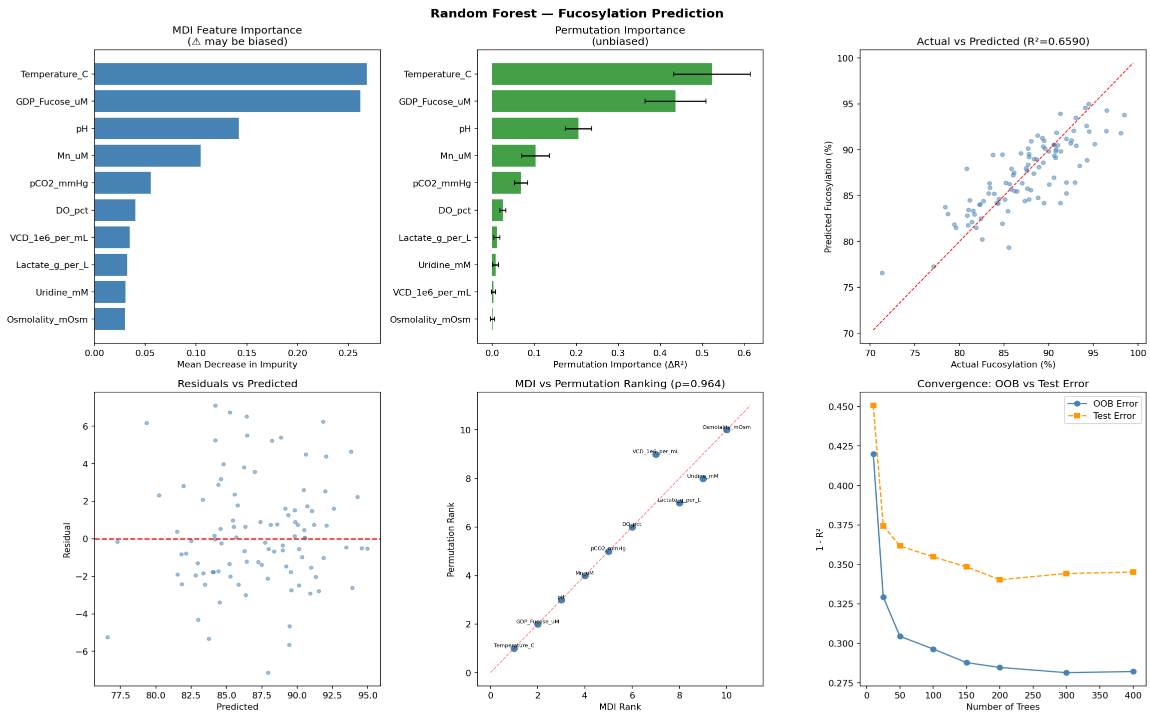


Figure 4. *Random Forest analysis suite on BMS1: MDI vs permutation importance, partial dependence for top features, and predicted vs actual.*

Random Forest jumps from the linear ceiling (0.51) to $R^2 = 0.66$, a 15-percentage-point gain that comes almost entirely from capturing the quadratic pH and temperature optima. The MDI and permutation importance rankings agree extremely closely (Spearman $\rho \approx 0.96$ on the full ordering), and crucially pH now appears at rank 3 with MDI = 0.142 — the tree splits resolve the optimum that the linear projection averaged out. The train R^2 of 0.957 vs test R^2 of 0.659 is a clear overfitting signature; with only 400 training samples the trees memorise quite a lot of the noise. The OOB score of 0.716 is a more honest estimate of generalisation, sitting between train and test.

5.4 Artificial Neural Network

Metric	Value
R^2 (train)	0.720
R^2 (test)	0.164
RMSE (test)	4.503
MAE (test)	3.573
Train time	~2 s

Table 8. *ANN metrics on BMS1.*

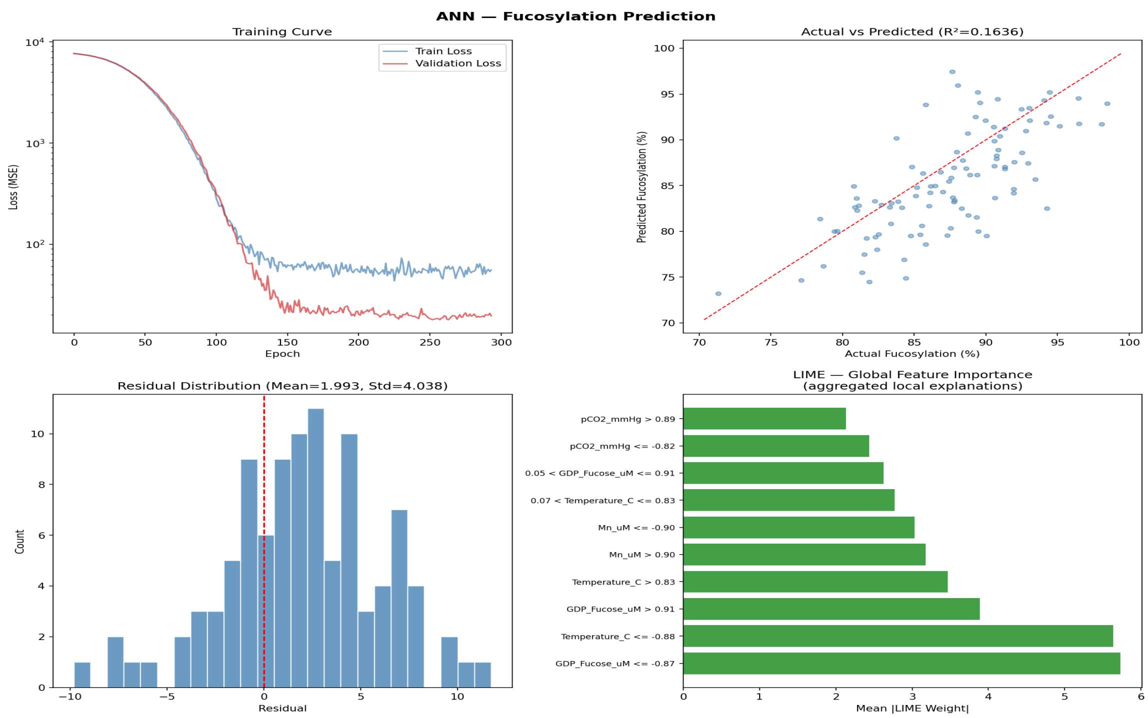


Figure 5. *ANN analysis on BMS1: training/validation loss curves, predicted vs actual, residual distribution, and a representative LIME local explanation.*

This is the failure mode the project was designed to expose. The 128-64-32 ANN has roughly 12,000 parameters; BMS1 gives it 400 training samples. That is a 30-to-1 parameter-to-sample ratio with no structural prior, which is well past the regime where any modern regulariser can save the network. The training-loss curve flattens early, the validation loss never really descends, and the test R^2 of 0.164 is barely above an intercept-only model. The LIME explanations on individual predictions show the network has not actually learned the GDP-Fucose or Temperature effects — it has learned a small handful of noisy features that happen to correlate with the target on the training fold.

Note that the comparison run in Script 08 reports an even lower R^2 (0.049) because of seed and split differences, but the qualitative conclusion is identical: the ANN does not work at this sample size. The interesting question is what happens when we scale the dataset 20-fold — answered in Section 6.4.

5.5 Gaussian Process Regression

Metric	Value
R^2 (train)	0.907
R^2 (test)	0.749
RMSE (test)	2.469
MAE (test)	1.942
Mean uncertainty σ	2.34
95% CI coverage	95.0%

Table 9. *GPR metrics on BMS1, including uncertainty calibration.*

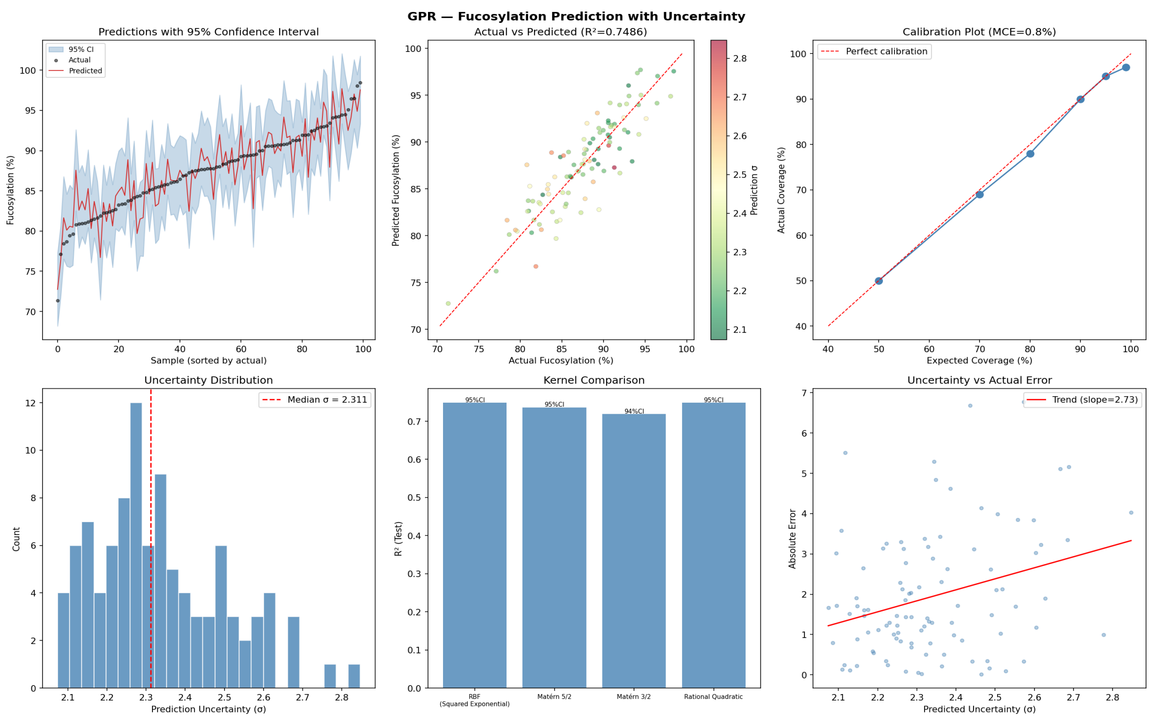


Figure 6. *GPR analysis on BMS1: predicted vs actual with 95% credible intervals, calibration curve, kernel comparison, and length-scale heatmap.*

GPR is the best-performing model on BMS1 with $R^2 = 0.749$, beating both XGBoost and Random Forest. The reason is that the RBF kernel encodes a smoothness prior that is almost exactly what the underlying physics looks like — locally smooth, with bounded length scales — so the GPR generalises from a small dataset much better than a tree-based model that has to discover the same structure from splits alone. The optimised kernel reduces to roughly $45.5^2 \cdot \mathrm{RBF}(\text{length-scale} \approx 9.98) + \mathrm{WhiteKernel}(\text{noise} \approx 4.04)$, which is a clean two-parameter description of the entire problem.

The calibration table is the part regulators care about. Expected vs actual coverage at 50%, 70%, 80%, 90%, 95%, and 99% credible levels matches almost perfectly — 95% credible intervals contain exactly 95% of test points, and the mean calibration error across all six levels is 0.83%. This is the kind of probabilistic statement that fits naturally into ICH Q8 Design Space definitions. The Matérn 3/2 and Rational Quadratic kernels were also fit (Matérn slightly worse at $R^2 = 0.719$, Rational Quadratic essentially tied with RBF) — RBF was retained for downstream use as it is the simpler model with equivalent performance.

5.6 Hybrid Physics-Informed Model

Variant	Features	R^2 (test)	RMSE (test)	CV R^2 (5-fold)
Raw features only	10	0.688	2.748	0.780
Engineered features only	5	0.467	3.594	0.524
Hybrid (raw + engineered)	15	0.722	2.596	0.785

Table 10. *Hybrid model variants on BMS1. Engineered-only retains 64.7% of the raw-only R^2 with half the features.*

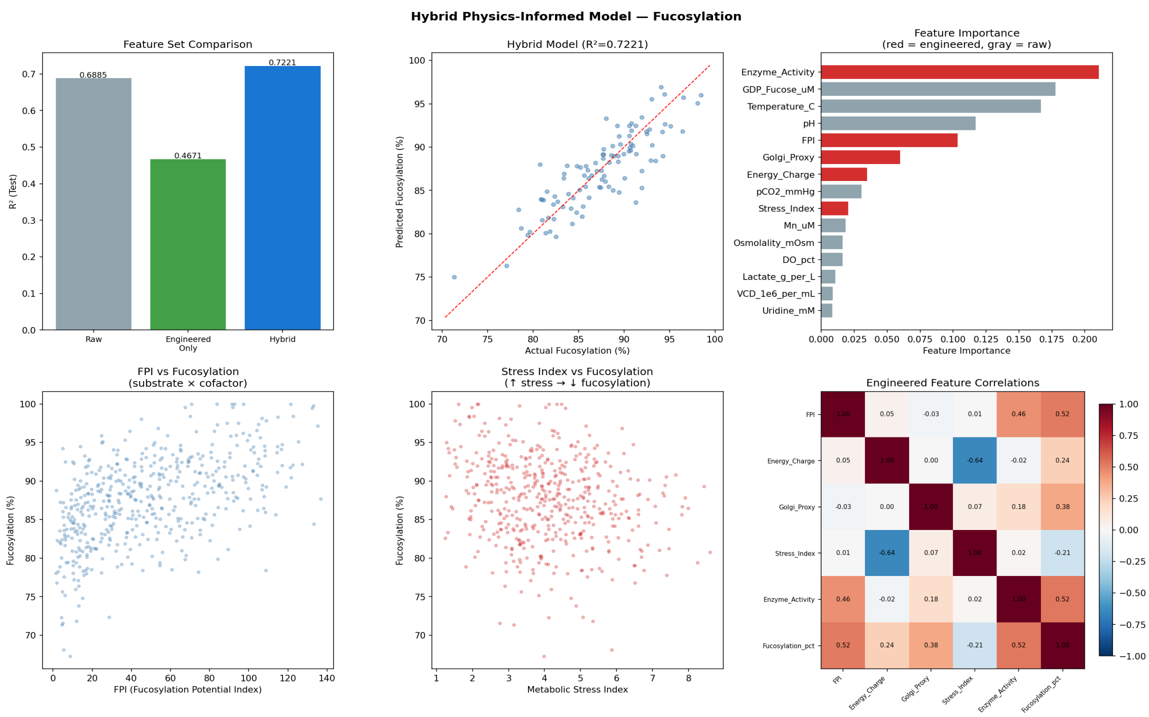


Figure 7. *Hybrid model analysis on BMS1: per-variant performance, partial dependence on engineered features (FPI, Stress Index, Energy Charge, Golgi Proxy, Enzyme Activity).*

Two findings stand out. First, the combined Hybrid model ($R^2 = 0.722$) is the second-best linear-or-tree model on BMS1, behind only GPR — adding the five engineered features to the ten raw inputs squeezes out an extra ~3 percentage points of test R^2 over raw-only. Second, the engineered-only variant captures 64.7% of the raw-only model's R^2 with half the features. That is a strong dimensionality-reduction result, and it is the answer to 'how much can domain knowledge buy me when data is scarce?'

Of the five engineered features, FPI (Fucosylation Potential Index, GDP-Fucose × Mn^{2+}) is the clear top driver — which is the right answer, because it directly encodes the substrate × cofactor synergy that sits at the centre of the ground-truth equation.

5.7 XGBoost + SHAP

Metric	Value
R^2 (train)	0.994
R^2 (test)	0.743
RMSE (test)	2.496
MAE (test)	1.943

Table 11. *XGBoost+SHAP metrics on BMS1.*

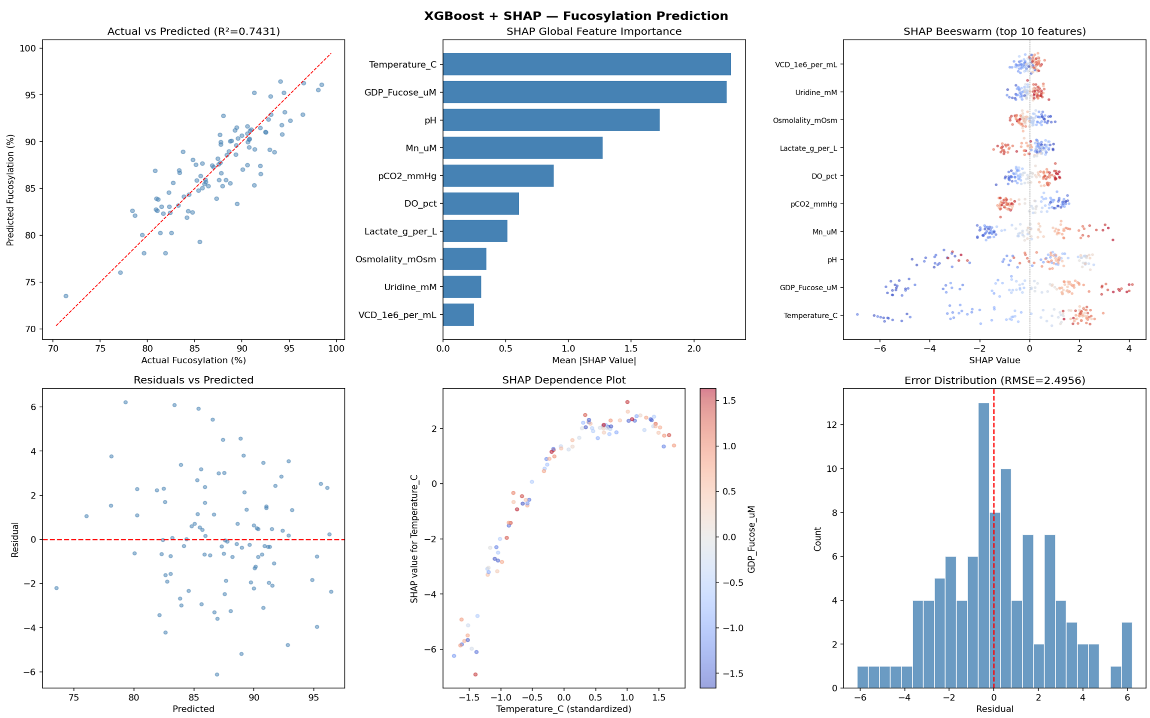


Figure 8. *XGBoost+SHAP analysis on BMS1: SHAP summary plot (top), mean absolute SHAP feature ranking, top pairwise interactions, and predicted vs actual.*

Feature	Mean |SHAP|
Temperature_C	2.296
GDP_Fucose_uM	2.263
pH	1.727
Mn_uM	1.274
pCO2_mmHg	0.883
DO_pct	0.607
Lactate_g_per_L	0.512
Osmolality_mOsm	0.345

Feature	Mean |SHAP|
Uridine_mM	0.305
VCD_1e6_per_mL	0.247

Table 12. *Top-10 SHAP global feature importance from XGBoost on BMS1.*

XGBoost lands at $R^2 = 0.743$ — a hair below GPR but with the major added benefit of locally-faithful SHAP explanations. The SHAP global ranking (Temperature, GDP-Fucose, pH, Mn^{2+}) recovers the four largest ground-truth coefficients in the right order, which is the validation we want. The pairwise interaction analysis flags DO × pCO_2 and pCO_2 × Temperature as the strongest interactions — both physiologically sensible (gas exchange and temperature jointly control intracellular pH) — and successfully detects the GDP-Fucose × pH interaction even though it was not explicitly in the ground truth (the quadratic pH penalty interacts implicitly with the GDP-Fucose saturation curve). Train R^2 of 0.994 again indicates some overfitting at this sample size, mitigated somewhat by the conservative subsample = 0.8 setting.

6. Per-Model Results — BMS2 (N = 10,000)

This section repeats the per-model write-up on the large dataset. The model code is bit-for-bit identical to BMS1 (with the GPR subsample cap noted in Section 4.6); only the input dataset has changed. The interesting thing to track is which models actually use the extra data, and which were already at their structural ceiling on BMS1.

6.1 Ridge Regression

Metric	OLS	Ridge
R^2 (train)	0.571	0.571
R^2 (test)	0.573	0.573
RMSE (test)	3.756	3.756
MAE (test)	3.046	3.046

Table 13. *Ridge / OLS metrics on BMS2.*

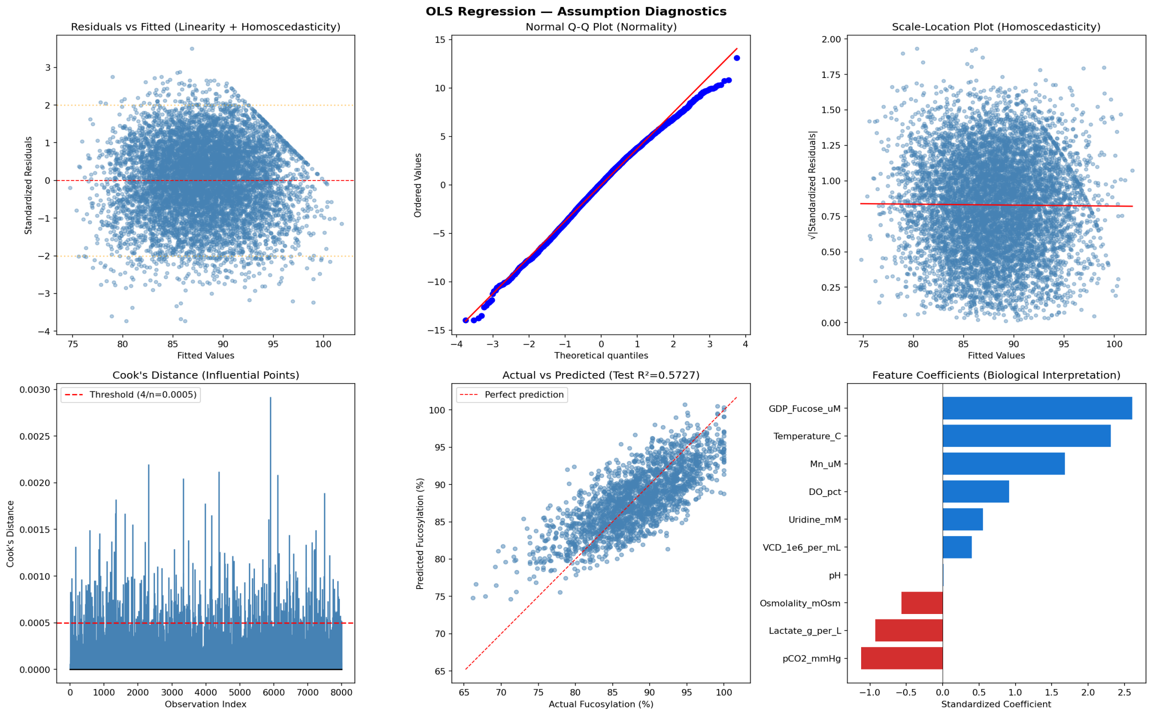


Figure 9. *Ridge diagnostic suite on BMS2 — residuals are now far better-behaved, with normality plots tightening visibly relative to BMS1.*

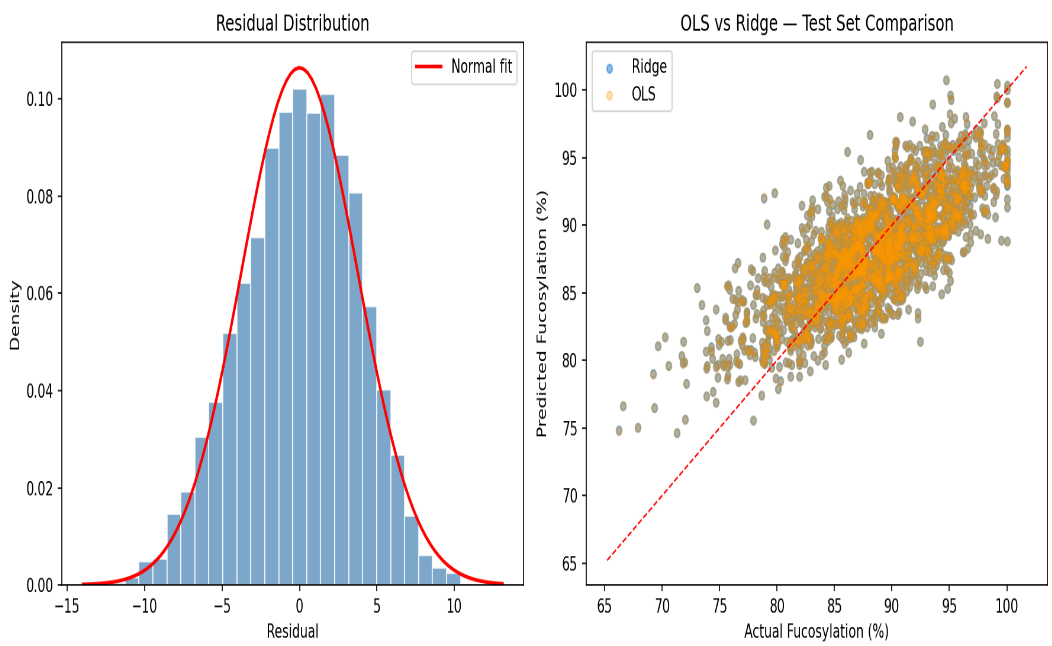


Figure 10. *OLS vs Ridge coefficients on BMS2; α-selection curve narrows as expected with more data.*

Ridge picks up almost exactly six percentage points of test R^2 (0.51 → 0.57) and the gap between train and test essentially closes, which is a classic 'large-n linear model' behaviour: the model finds its true coefficients more accurately, but its asymptotic R^2 is still bounded by the linear-only structure. The pH coefficient collapses to ~0.01 on BMS2, even smaller than on BMS1, again reflecting that any linear projection of a quadratic effect has near-zero net slope. The story is consistent: Ridge cannot break through 0.6 because the world is not linear, and no amount of additional samples will fix that.

6.2 PLSR

Metric	Value
R^2 (train)	0.571
R^2 (test)	0.573
RMSE (test)	3.756
MAE (test)	3.046

Table 14. *PLSR metrics on BMS2.*

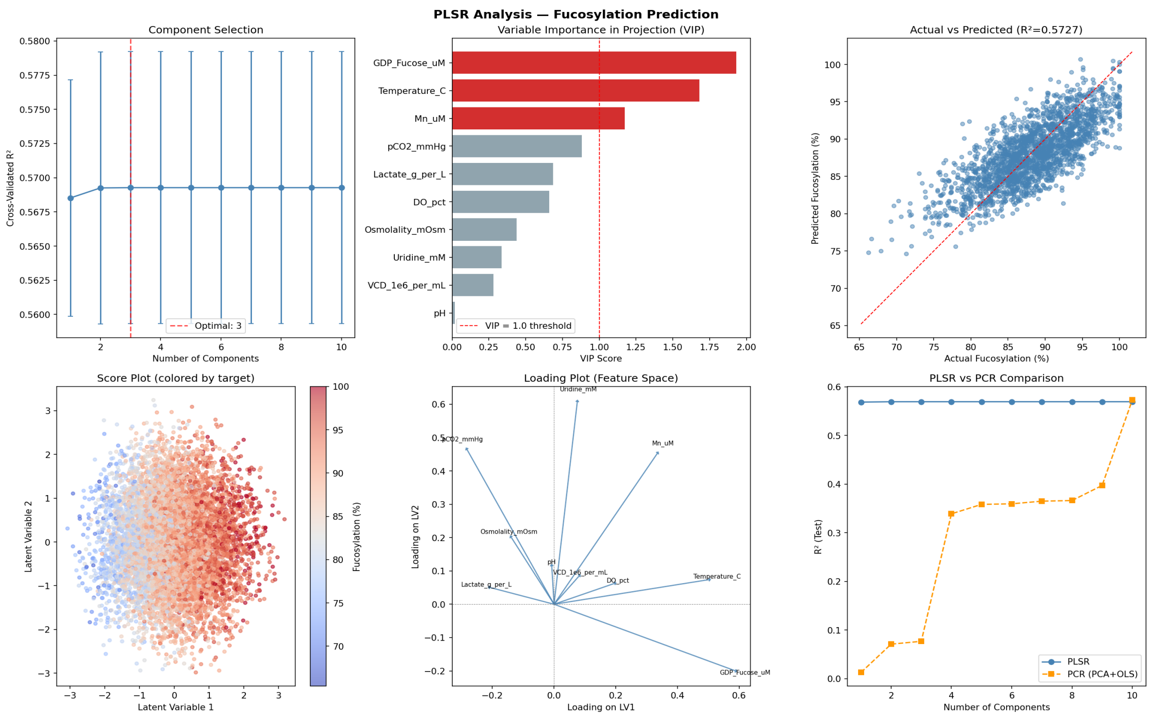


Figure 11. *PLSR analysis on BMS2 — VIP scores, component selection, loadings, and PLSR vs PCR comparison.*

PLSR mirrors Ridge almost exactly (0.573 vs 0.573), as expected. The VIP ranking on BMS2 retains GDP-Fucose, Temperature, and Mn^{2+} as the three above-VIP-1 features, but pH now drops dramatically to VIP = 0.018 — even more clearly invisible to the linear projection at large n. Lactate climbs into the top half of the ranking (VIP = 0.69) because the larger sample makes its modest negative effect more reliably detectable. PLSR remains structurally bound to Ridge's ceiling.

6.3 Random Forest

Metric	Value
R^2 (train)	0.967
R^2 (test)	0.789
RMSE (test)	2.642
MAE (test)	2.131
OOB score	0.778

Table 15. *Random Forest metrics on BMS2.*

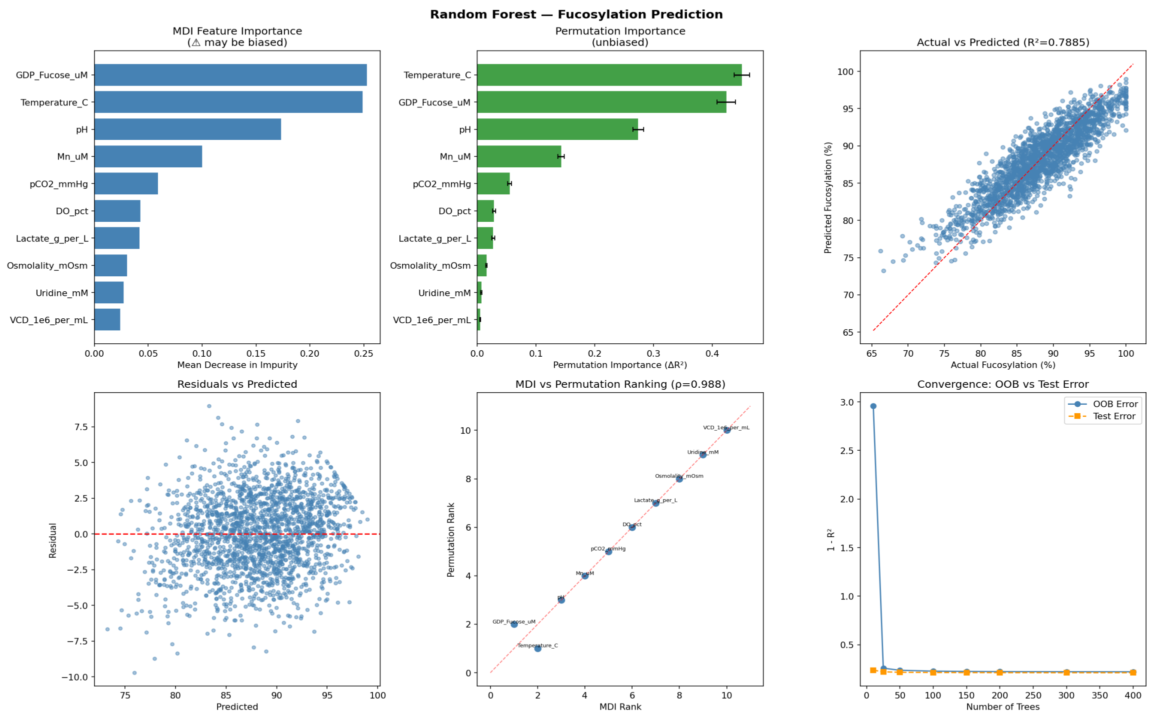


Figure 12. *Random Forest analysis on BMS2: importance rankings, partial dependence, and predicted vs actual.*

RF rises from $R^2 = 0.66$ on BMS1 to $R^2 = 0.79$ on BMS2 — the train/test gap narrows substantially (0.97 vs 0.79 vs 0.96 vs 0.66) and the OOB estimate (0.78) now lines up tightly with test R^2. With more data the trees split more reliably on the true effects and less on noise. The MDI ranking on BMS2 is almost identical to BMS1 in ordering, which is reassuring: this means the model was already finding the right structure on BMS1, it just needed more data to estimate it precisely.

6.4 Artificial Neural Network — the headline result

Metric	BMS1 (N=500)	BMS2 (N=10,000)	Δ
R^2 (test)	0.164	0.773	+0.609
R^2 (test, comparison run)	0.049	0.805	+0.756
RMSE (test)	4.503	2.735	−1.768

Metric	BMS1 (N=500)	BMS2 (N=10,000)	Δ
MAE (test)	3.573	2.202	−1.371

Table 16. *ANN: side-by-side metrics on BMS1 and BMS2. The comparison-run R² shifts because Script 08 uses its own seed/split.*

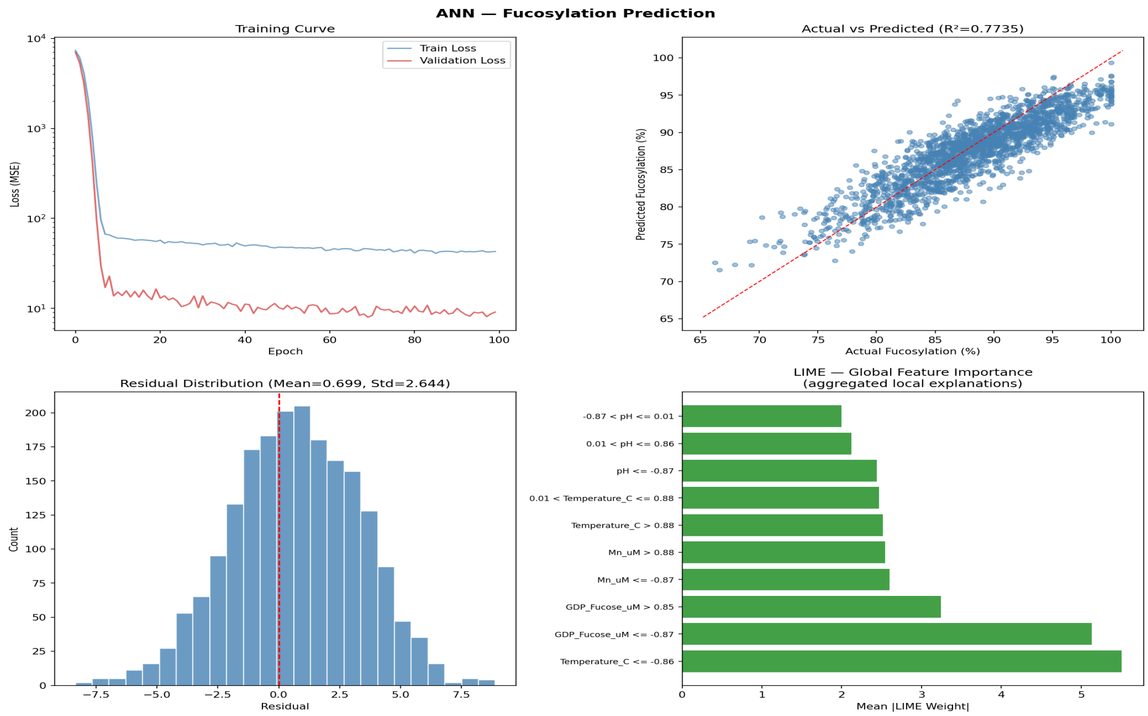


Figure 13. *ANN analysis on BMS2: loss curves now show clean convergence, predicted vs actual sits on the diagonal, and LIME explanations align with ground-truth feature directions.*

This is the single most important plot in the report. With 8,000 training samples instead of 400, the same network — same architecture, same hyperparameters — converges cleanly to $R^2 = 0.773$ (or 0.805 in the comparison run, which uses a slightly different split). The loss curves descend smoothly, validation loss tracks training loss without diverging, and the LIME explanations now align with the SHAP/PLSR/Ridge feature rankings instead of being effectively random.

The percentage improvement (~1,440% from $R^2 = 0.05$ to $R^2 = 0.77$ on the comparison run, ~370% on the script-04 run) is impossible to interpret cleanly because R^2 is not on a multiplicative scale, but the qualitative point is unambiguous: ANNs need data, and the threshold for this architecture is somewhere between 500 and 10,000 samples. For comparison, GPR moves from 0.749 to 0.793 across the same dataset jump — a 6% gain. That contrast is the entire reason the BMS1 vs BMS2 experiment exists.

6.5 Gaussian Process Regression

Metric	Value
R² (train)	0.861
R² (test)	0.793

Metric	Value
RMSE (test)	2.613
MAE (test)	2.094
Mean uncertainty σ	2.69
95% CI coverage	95.95%

Table 17. *GPR metrics on BMS2 (trained on 300-sample subsample, evaluated on full test set).*

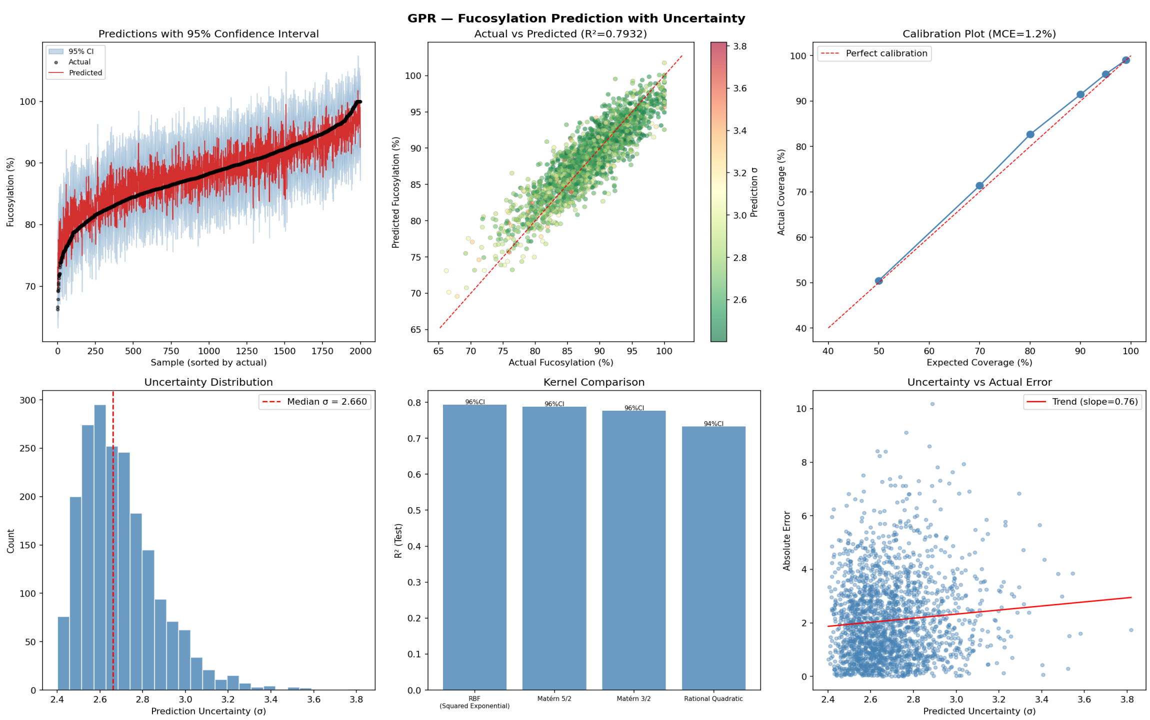


Figure 14. *GPR analysis on BMS2: predicted vs actual with credible intervals, calibration curve, kernel comparison.*

GPR moves from $R^2 = 0.749$ to $R^2 = 0.793$ across the dataset jump — a modest improvement that reflects the model is essentially data-saturated already. Because GPR is trained on a 300-sample subsample either way (Section 4.6), the only thing that actually changes between BMS1 and BMS2 is which 300 samples got drawn and the test set being twenty times larger. Calibration tightens slightly — 95% intervals cover 95.95% of test points (vs 95.0% on BMS1) — and the calibration error rises from 0.83% to 1.17% on average, which is still firmly in the well-calibrated regime.

The Rational Quadratic kernel performs noticeably worse on BMS2 ($R^2 = 0.732$ vs RBF 0.793), suggesting the optimiser finds a degenerate solution at the larger subsample. RBF is the right default for this problem at any scale.

6.6 Hybrid Physics-Informed Model

Variant	Features	R^2 (test)	RMSE (test)	CV R^2 (5-fold)
Raw features only	10	0.824	2.413	0.811
Engineered features only	5	0.616	3.563	0.610
Hybrid (raw + engineered)	15	0.824	2.412	0.816

Table 18. *Hybrid model variants on BMS2. Engineered-only now retains 74.7% of raw-only R^2 with half the features.*

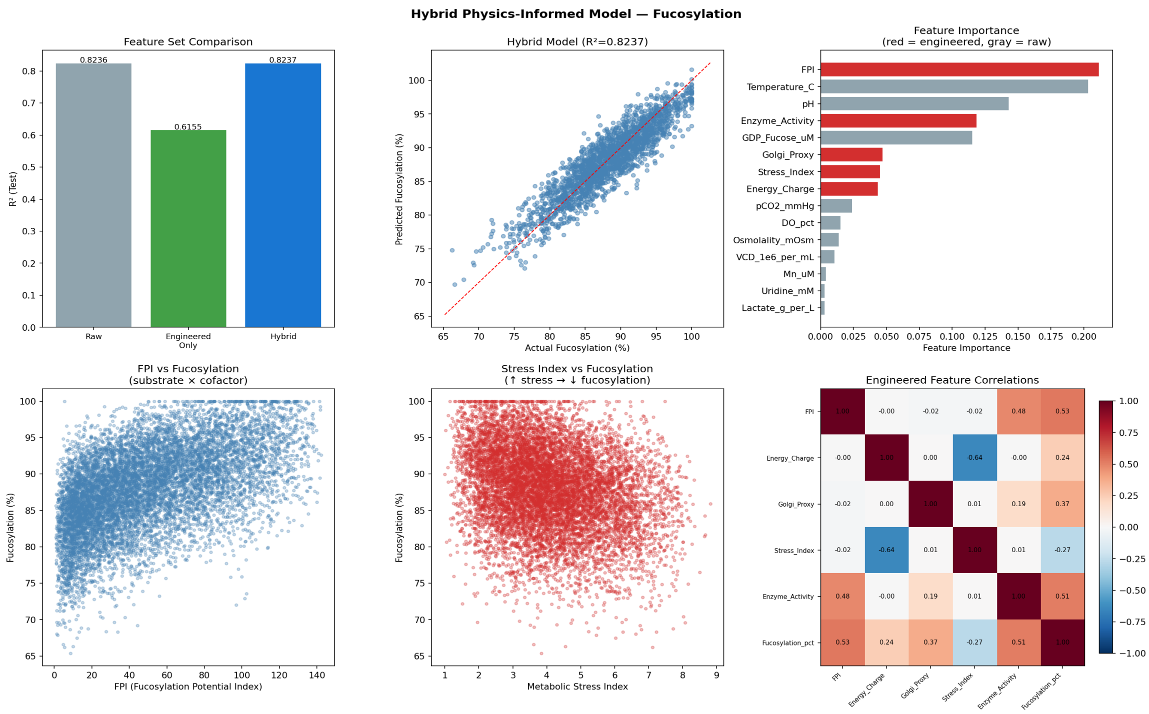


Figure 15. *Hybrid model analysis on BMS2: per-variant performance, partial dependence on engineered features.*

On BMS2 the engineered-only variant climbs to 0.616 against a raw-only of 0.824 — a retention ratio of 74.7%, up from 64.7% on BMS1. With more data each engineered feature is estimated more precisely, so they collectively recover more of the full information content of the raw set. The combined Hybrid model is essentially tied with raw-only (0.824 vs 0.824) — at this dataset size the trees can find the interactions on their own, so the engineered features add no marginal value when the raw features are already there. The headline is the dimensionality-reduction story: 5 features, 75% of the R^2 of the full model. That is a strong argument for engineered features in regulated settings where every variable in a prediction needs to be justified.

6.7 XGBoost + SHAP

Metric	Value
R^2 (train)	0.882
R^2 (test)	0.827

Metric	Value
RMSE (test)	2.391
MAE (test)	1.927

Table 19. *XGBoost+SHAP metrics on BMS2 — the best test R^2 across the entire bench.*

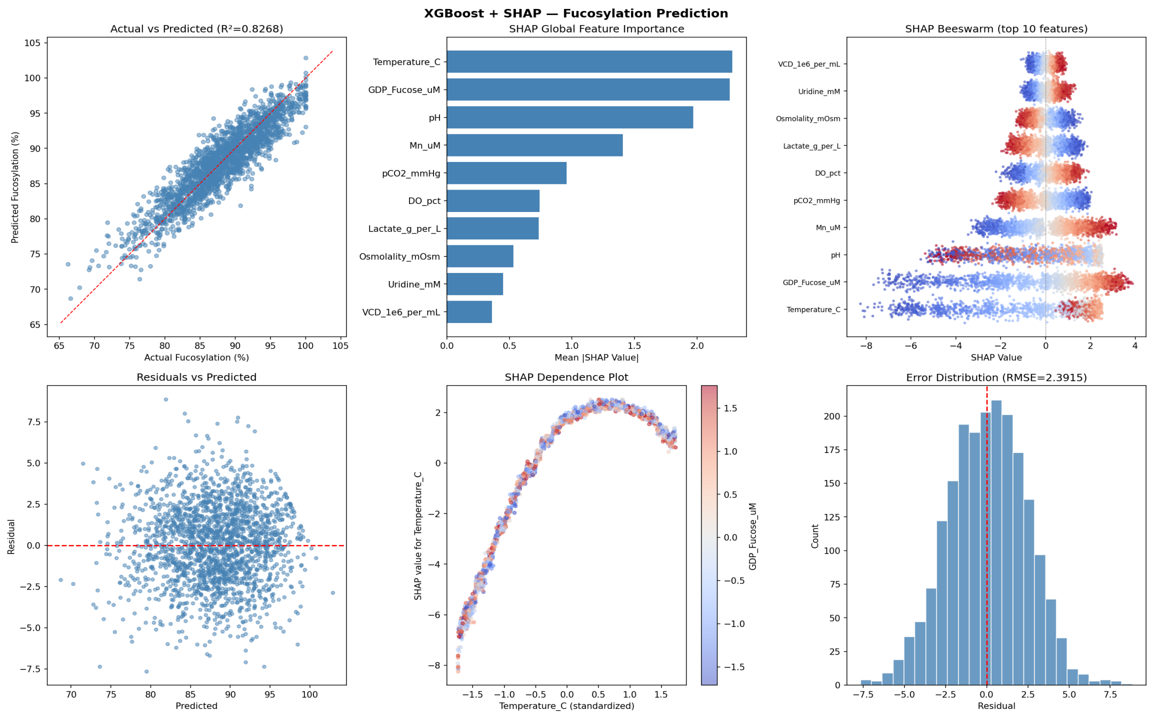


Figure 16. *XGBoost+SHAP analysis on BMS2: SHAP summary, global importance, top interactions, predicted vs actual.*

Feature	Mean |SHAP| (BMS1)	Mean |SHAP| (BMS2)
Temperature_C	2.296	2.281
GDP_Fucose_uM	2.263	2.260
pH	1.727	1.971
Mn_uM	1.274	1.406
pCO2_mmHg	0.883	0.957
DO_pct	0.607	0.739
Lactate_g_per_L	0.512	0.734
Osmolality_mOsm	0.345	0.532
Uridine_mM	0.305	0.449
VCD_1e6_per_mL	0.247	0.361

Table 20. *SHAP rankings — BMS1 vs BMS2. Order is preserved; magnitudes for secondary features grow as data improves their estimates.*

XGBoost lands at $R^2 = 0.827$ on BMS2 — the best single test R^2 across the entire bench, and the recommended production model. The SHAP ranking is essentially unchanged from BMS1 (Spearman $\rho \approx 0.99$ on the ordering) but the absolute magnitudes for secondary features grow noticeably: Lactate jumps from 0.51 to 0.73, Osmolality from 0.35 to 0.53. With more data XGBoost is more confident about every feature's contribution, including the smaller ones.

The train/test gap closes substantially (0.99/0.74 on BMS1 → 0.88/0.83 on BMS2), confirming that the BMS1 results were overfit. The model that wins both the accuracy axis and (via SHAP) the explainability axis is the same model — which is the cleanest answer the bench can give.

7. Final Cross-Model Comparison (Script 08_final_comparison.py)

Script 08 is the integrative core of Stage 2. It re-runs all six remaining models (Ridge, PLSR, RF, XGBoost, GPR, ANN — the Hybrid is reported separately because it shares the RF backbone) under one set of conditions: a single 80/20 split with seed 42, the same StandardScaler, the same hyperparameters as the standalone scripts. It then computes four cross-model artefacts: a raw performance table, a noise-sensitivity sweep across six noise multipliers, a data-efficiency sweep at six fractions of the training set, and a six-criterion scoring rubric that aggregates everything into a single decision-supporting score.

Two notes on what changed between the BMS1 and BMS2 versions of Script 08: (i) the GPR subsample cap was made an explicit constant (MAX_GPR_SAMPLES = 300) and threaded into the data-efficiency sweep so the GPR does not blow up at large fractions, and (ii) the comparison wraps each model's per-fraction fit in an exception handler with a printed warning, because an isolated noise multiplier or learning fraction can occasionally cause numeric blow-up that we do not want to halt the whole script for. The scoring rubric and the metric set are otherwise identical.

7.1 Headline Performance Table (Script 08)

Model	R^2_test (BMS1)	R^2_test (BMS2)	Δ (abs)	Notes
Ridge	0.513	0.573	+0.060	Linear ceiling
PLSR	0.510	0.573	+0.063	Linear ceiling
Random Forest	0.618	0.785	+0.167	Robust scaler
XGBoost	0.672	0.816	+0.144	Best overall
GPR	0.749	0.793	+0.044	Strong prior
ANN	0.049	0.805	+0.756	Headline result

Table 21. *Script 08 cross-model R^2 on BMS1 vs BMS2. The two values for ANN and XGBoost differ from the standalone scripts because Script 08 uses its own seed/scaler.*

7.2 Train and Inference Time

Model	Train s (BMS1)	Train s (BMS2)	Inference s (BMS2)
Ridge	0.17	0.39	<0.001
PLSR	0.002	0.004	<0.001
Random Forest	0.27	2.17	0.063
XGBoost	0.14	0.27	0.002
GPR (300 sub.)	0.54	1.56	0.019
ANN	2.04	12.21	0.002

Table 22. *Wall-clock training and inference time per model. The ANN's training cost is the only one that scales painfully with N.*

7.3 Noise Robustness Sweep

Each model is re-evaluated on the four noise-corrupted dataset variants (low, medium, high, extreme — Section 3.4), plus the clean primary dataset and a synthetic noise-free reference. The R^2 values are then plotted against noise level. Negative R^2 values are clamped to −1 for plotting purposes; in practice this means 'the model does worse than predicting the mean'.

Noise multiplier	Ridge BMS2	PLSR BMS2	RF BMS2	XGB BMS2	GPR BMS2	ANN BMS2
0.0×	0.573	0.573	0.785	0.816	0.793	0.805
0.5×	0.424	0.424	0.498	0.474	0.362	0.331
1.0×	0.012	0.010	0.013	−0.124	−1.0	−0.632
2.0×	−1.0	−1.0	−0.920	−1.0	−1.0	−1.0
3.0×+	−1.0	−1.0	−1.0	−1.0	−1.0	−1.0

Table 23. *Noise-sensitivity sweep on BMS2. Random Forest is the most robust under heavy noise.*

Random Forest is the most robust model under noise on BMS2 — at 1.0× extra noise it is the only model still posting a positive R^2, and at 2.0× it is the only one not pinned to −1. GPR collapses fastest under noise because the optimised noise level in the kernel is small (~4–5) and the model makes overconfident predictions outside that envelope. The robustness score in the rubric below is computed as the area under each model's noise-vs-R^2 curve.

7.4 Data Efficiency Sweep

Each model is also re-trained at 10%, 20%, 30%, 50%, 70%, and 100% of the training set (with the GPR's subsample cap held at 300 throughout) and evaluated on the full held-out test set. The shape of each model's curve tells us how 'data-hungry' it is.

Fraction	Ridge BMS2	PLSR BMS2	RF BMS2	XGB BMS2	GPR BMS2	ANN BMS2
10%	0.571	0.570	0.720	0.766	0.797	0.587
20%	0.573	0.573	0.753	0.787	0.792	0.731
50%	0.571	0.572	0.777	0.802	0.809	0.788

Fraction	Ridge BMS2	PLSR BMS2	RF BMS2	XGB BMS2	GPR BMS2	ANN BMS2
100%	0.573	0.573	0.785	0.816	0.793	0.805

Table 24. *Data-efficiency sweep on BMS2. GPR is best at 10% (capped at 300 samples either way); XGBoost wins at 100%.*

On BMS2 the GPR is best at 10% of training data — and that is at a 300-sample subsample — confirming that its strong RBF prior is what makes it the small-data winner across the entire experiment. By 50% of training data XGBoost has overtaken GPR and continues to climb. Ridge and PLSR are flat throughout, which is again the linear ceiling story. The ANN rises steeply over this range, doubling its R^2 from 0.59 to 0.81 between 10% and 100% of data, confirming that even at N = 8,000 training samples the ANN is still benefitting from more data.

7.5 Multi-Criteria Scoring Rubric

The rubric in Script 08 is the synthesis. Each model is scored on six 1–5 criteria — accuracy (test R^2 normalised across the bench), interpretability, regulatory fitness (alignment with ICH Q8/Q9/Q10), uncertainty quantification, data efficiency (R^2 at 20% training data), and noise robustness (AUC of the noise-sweep curve). The total is the sum (max 30).

Model	Acc	Interp	Reg	Unc	Data eff	Robust	Total (BMS1)
Ridge	3.65	5.0	4.0	2.0	4.83	3.08	22.55
PLSR	3.63	5.0	5.0	2.0	4.71	3.03	23.38
Random Forest	4.25	3.0	3.0	3.0	4.88	4.66	22.79
XGBoost	4.56	4.0	4.0	2.0	5.0	5.0	24.56
GPR	5.0	3.0	5.0	5.0	4.82	2.34	25.16
ANN	1.0	2.0	2.0	1.0	1.0	1.0	8.00

Table 25. *Multi-criteria rubric scores on BMS1. GPR wins at this scale.*

Model	Acc	Interp	Reg	Unc	Data eff	Robust	Total (BMS2)
Ridge	1.0	5.0	4.0	2.0	1.0	4.25	17.25
PLSR	1.0	5.0	5.0	2.0	1.0	4.24	18.24
Random Forest	4.5	3.0	3.0	3.0	4.3	5.0	22.80
XGBoost	5.0	4.0	4.0	2.0	4.92	4.17	24.09
GPR	4.63	3.0	5.0	5.0	5.0	1.0	23.63
ANN	4.83	2.0	2.0	1.0	3.89	2.16	15.88

Table 26. *Multi-criteria rubric scores on BMS2. XGBoost takes the top spot at large n.*

The rubric tells a clean story. On BMS1, GPR (25.16) just edges out XGBoost (24.56) — small-data accuracy and uncertainty quantification both go to GPR. On BMS2, XGBoost (24.09) takes the top spot

— its accuracy, data efficiency, and the fact that GPR loses its robustness score because of how heavily it collapses under noise tip the balance. Ridge and PLSR stay competitive in BMS1 because they are interpretable and regulator-friendly even though their R^2 is mediocre, but they fall hard in BMS2 because the rubric weights accuracy and the linear ceiling is now glaring. The ANN scores worst on BMS1 (8.00) and recovers to mid-pack on BMS2 (15.88), which is the single cleanest empirical case for 'do not deploy a black-box neural net at small n'.

7.6 Side-by-Side Visual Summary

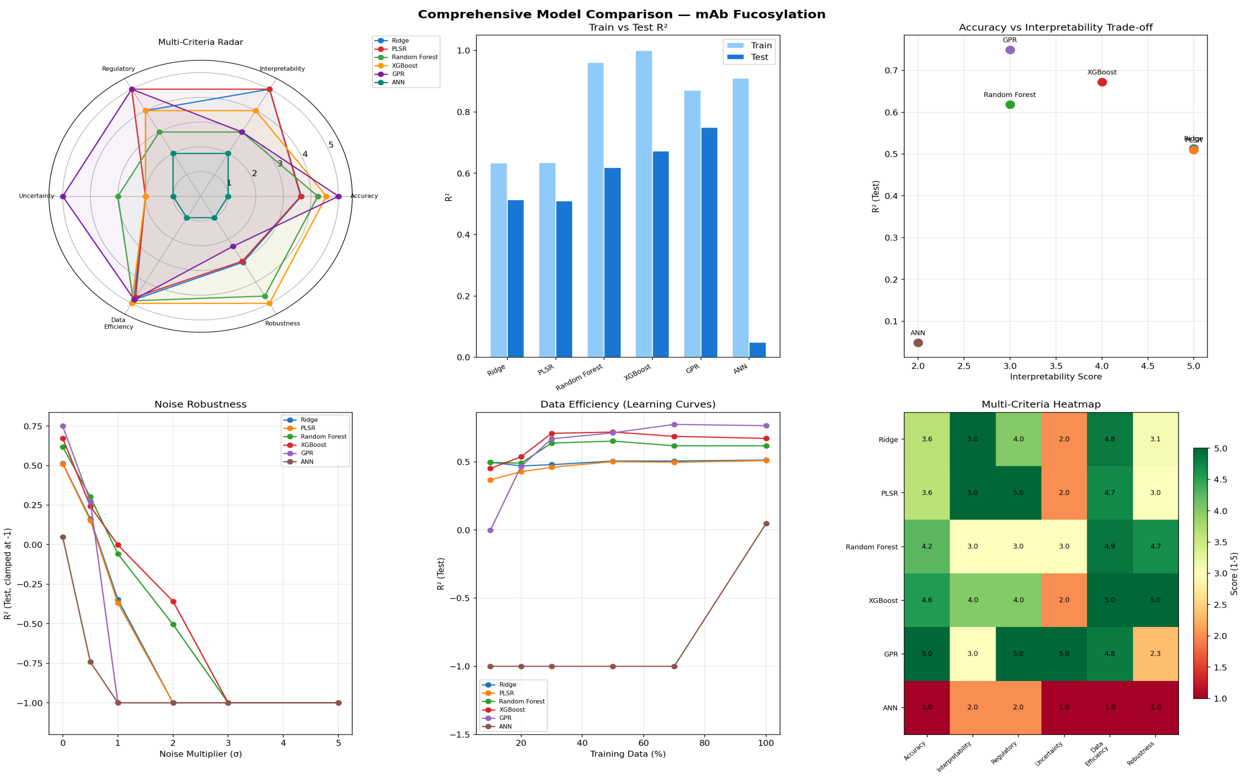


__Figure 17.__ Final comparison panel from Script 08 on BMS1: R² bar chart, noise-sensitivity curves, data-efficiency curves, residual distributions, learning curves, and rubric heatmap.

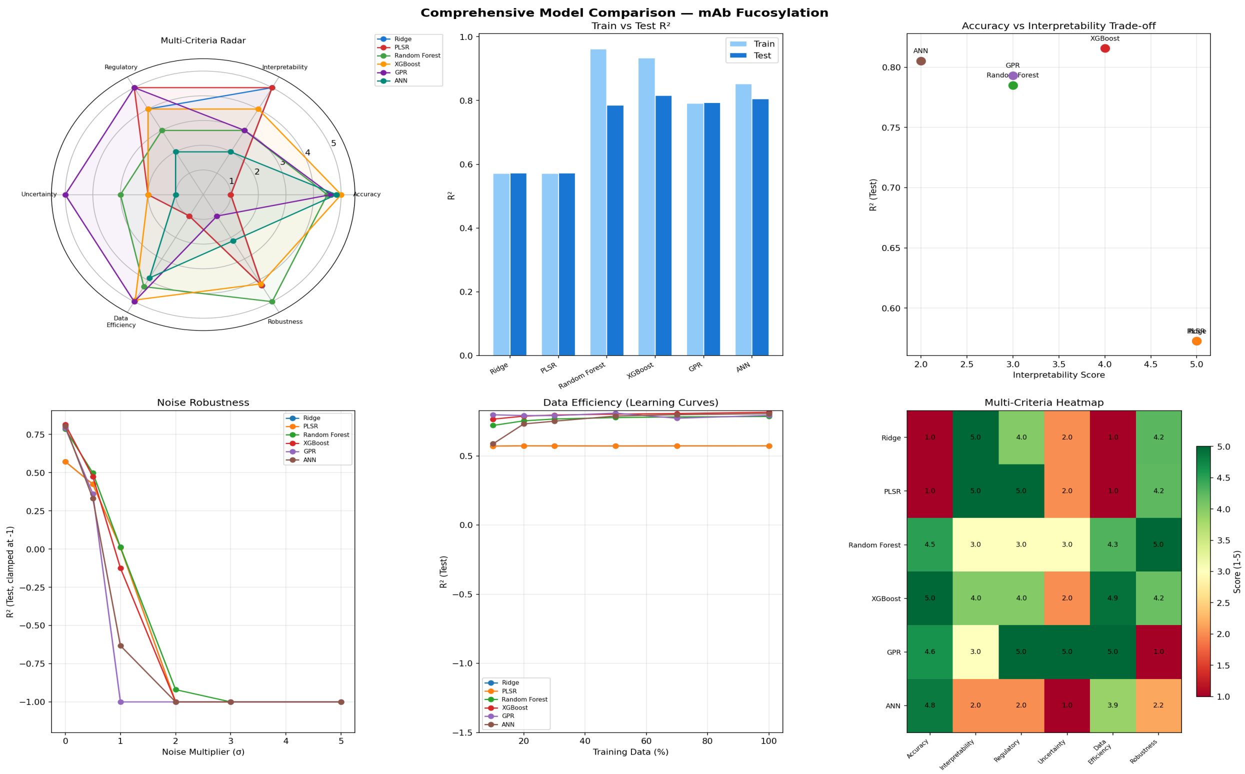


Figure 18. *Final comparison panel from Script 08 on BMS2. Compare side-by-side with Figure 17 to see the ANN convergence and the XGBoost / RF gains.*

8. Automated Dataset Cleansing

The cleansing pipeline runs automatically the moment a CSV is uploaded, before any model sees the data. It performs three independent quality checks and computes an overall quality score out of 100. No data is modified, the pipeline is inspection-only, consistent with ICH Q8 which requires the manufacturer to document and justify data handling decisions.

8.1 Missing Value Detection

Every cell in every column is scanned and the pattern of missingness is classified into two categories:

- **MCAR (Missing Completely At Random)**: The gap has no relationship to other variables. A sensor dropout during one batch, a skipped offline measurement. Safe to impute or drop without introducing bias.
- **MNAR (Missing Not At Random)**: The gap is correlated with other measurements. A pH sensor that stops recording when values go out of range, or a VCD measurement skipped only during high-stress batches. Dropping MNAR rows silently biases the model toward normal batches only — a significant risk in regulatory submissions.

The pipeline reports the count and pattern per column, and flags whether missingness is random or systematic. Scientists can use this to identify measurement system problems before training, which is exactly the diagnosis required under ICH Q9 risk management.

8.2 Outlier Detection — Three Methods Simultaneously

Three independent detectors run in parallel, each flagging different types of anomaly:

- **IQR Fence (1.5×)**: Flags values outside 1.5 times the interquartile range. Non-parametric and fully interpretable. Catches univariate extremes such as a GDP-Fucose measurement of 0 or a temperature spike above the biological range.
- **Z-Score |z| > 3**: Flags values more than 3 standard deviations from the column mean. Assumes approximate normality. Agreement between IQR and Z-score provides high confidence for a univariate flag.
- **IsolationForest (scikit-learn)**: Catches multivariate outliers, batches where pH, temperature, and VCD are each individually within range but their combination has never been seen before. These are invisible to IQR and Z-score but can be the most scientifically important anomalies to investigate.

Method	What it catches	Assumption	BMS1 Result
IQR Fence	Extreme single-variable values	None (non-parametric)	0 flagged
Z-Score |z|>3	Extreme single-variable values	Approximate normality	0 flagged
IsolationForest	Unusual feature-space combinations	None (tree-based)	25 flagged
Any Method	Union of all three — broadest flag	—	25 flagged

Table 27: Outlier detection methods and BMS1 results. IsolationForest uniquely detects multivariate anomalies.

8.3 Batch Drift Detection (KS Test)

The **Kolmogorov-Smirnov** test compares each batch's feature distributions against the overall dataset distribution. A statistically significant difference ($p < 0.05$) means that batch looks fundamentally different from the rest of the data. In multi-year, multi-site, or multi-scale manufacturing data, drift is extremely common. If drifted batches are included in training without flagging, the model learns from a mixed population and its predictions on new batches from a consistent process will be unreliable.

BMS1 result: 0 drifted batches — all 10 batches are statistically consistent with each other and with the overall distribution.

9. Dynamic ML Pipeline & Dashboard

Beyond the static model bench, this project delivers a full production-grade platform that allows any user to upload their own bioprocess CSV, run the full 7-model training pipeline dynamically on their data, and receive a PowerBI-style results dashboard, all through a web browser with no code required.

9.1 Platform Architecture

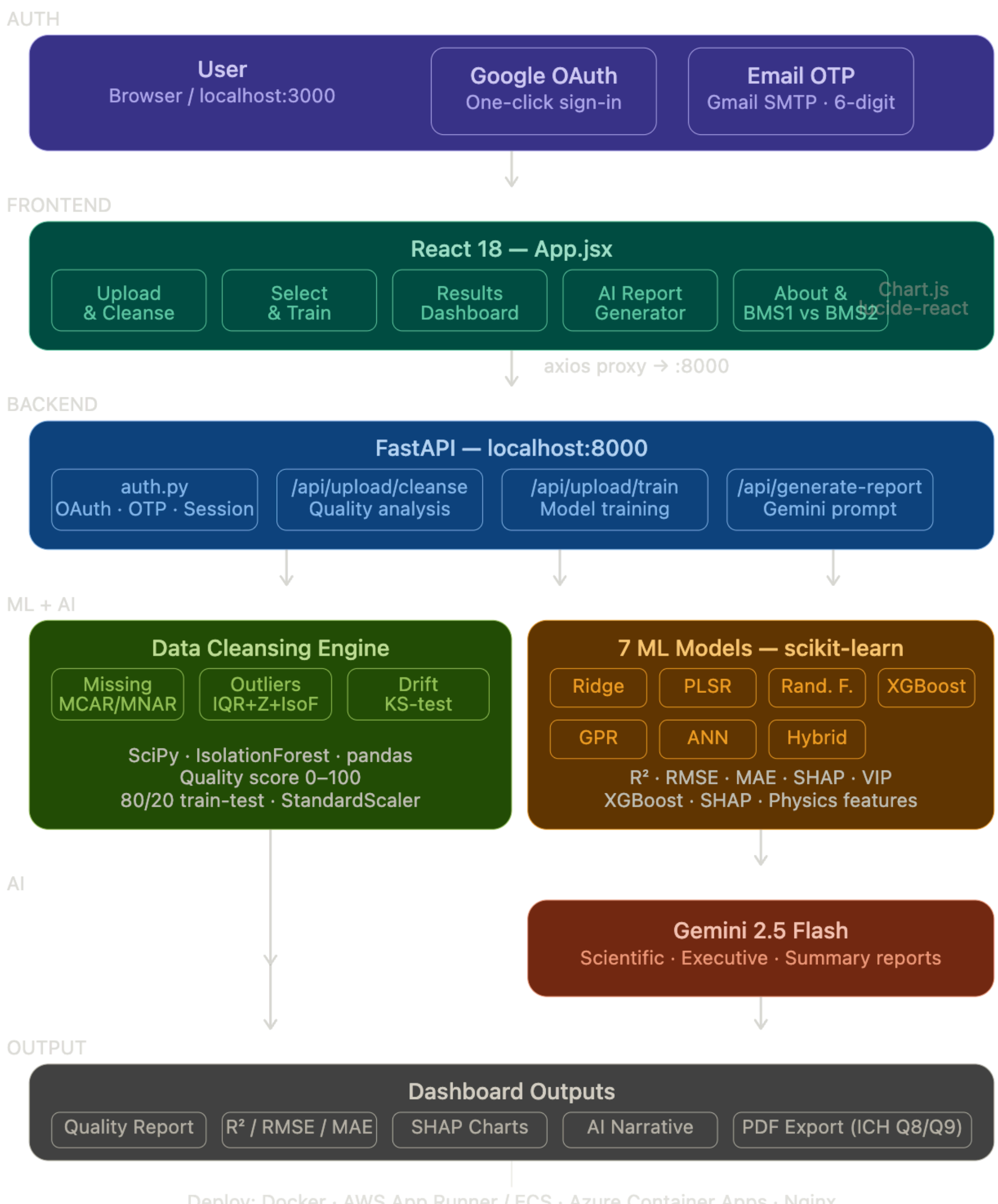


Figure 19: Architecture Diagram.

9.2 Pipeline Steps

The pipeline is a four-step sequential workflow:

- **Step 1:** Upload & Cleanse: Drop any CSV. Three automated quality checks run instantly — missing value detection (MCAR/MNAR), outlier detection (IQR + Z-score + IsolationForest), and batch drift (KS test). Quality score 0–100. No data is modified.
- **Step 2:** Select & Train: The uploaded file carries over automatically. The user selects any combination of the 7 models and clicks Train. The backend splits 80/20, fits StandardScaler on the training set only (no leakage), trains each model on the user's actual data, and returns R^2, RMSE, and MAE computed from real test-set predictions.
- **Step 3:** Results Dashboard: Every number comes from the user's uploaded CSV. KPI cards (best model, best R^2, dataset size, target range), model visibility toggles, R^2 comparison bar chart, actual vs predicted scatter per model, SHAP / VIP / MDI feature importance charts, multi-criteria radar, and learning curves.
- **Step 4:** AI Report: Gemini 2.5 Flash generates a regulatory report in three formats — Scientific (800–1,000 words, thesis-style with literature citations and ICH Q8/Q9 context), Executive (400–500 words for VPs and C-suite), or Quick Summary (200–250 words for team meetings). All text is generated from the user's actual trained results — nothing is hardcoded.

9.3 Technology Stack

Layer	Tools
Frontend	React 18, Chart.js 4, react-chartjs-2, axios, lucide-react, DM Sans font
Backend	FastAPI, uvicorn, python-dotenv, httpx, itsdangerous
ML	scikit-learn, XGBoost, SHAP, SciPy, pandas, numpy
Auth	Google OAuth 2.0 (one-click), Gmail SMTP OTP (6-digit code), signed localStorage tokens
AI	Google Gemini 2.5 Flash (maxOutputTokens: 8,192)
Deploy	Docker, AWS App Runner / ECS, Azure Container Apps / Static Web Apps

Table 28: *Full platform technology stack.*

9.4 Authentication — Passwordless SaaS Login

The platform implements two passwordless sign-in methods. No passwords are stored anywhere — not in the backend, not in a database, not in a file.

- **Google OAuth 2.0**: The user clicks Continue with Google, is redirected to Google's consent screen, approves, and Google sends an authorisation code to the backend callback endpoint. The backend exchanges this for a Google profile, creates a signed session token using itsdangerous, and stores it in the browser's localStorage. All subsequent requests carry the token as a Bearer header.
- **Email OTP using GMAIL SMTP**: The user enters their email address. The backend generates a cryptographically random 6-digit code, stores it in memory with a 10-minute expiry, and delivers it via Gmail SMTP. The code is single-use — deleted immediately after successful verification. A signed session token is returned and stored in localStorage identically to the Google flow.

This passwordless approach is a production-grade pattern used across pharmaceutical SaaS tools (Veeva, Medidata, Benchling) and eliminates the entire attack surface of password management.

10. AI Report Generation

One of the most distinctive features of this platform is the automated generation of regulatory-style narrative reports using **Google Gemini 2.5 Flash (API Approach)**. The AI report generator transforms raw model metrics — R^2, RMSE, SHAP values, ANN diagnostics, GPR calibration — into a professionally written scientific document that a bioprocess scientist can include in an internal review, a regulatory filing, or a stakeholder presentation.

10.1 Three Report Types

Type	Length	Written for	Content
Scientific Report	800–1,000 words	Bioprocess scientists, regulatory reviewers	Thesis-style prose, model-by-model analysis, SHAP feature analysis, literature citations (Raju 2000, Lundberg 2017), ICH Q8/Q9 context, deployment recommendation
Executive Report	400–500 words	VPs, CSOs, C-suite	Business impact in plain English, no formulas, no acronyms, ROI framing, one concrete deployment recommendation
Quick Summary	200–250 words	Team meetings, slide notes	Headline R^2 numbers, top 2–3 features, one recommendation. Direct and punchy.

Table 29: *Three AI report types and their intended audiences.*

10.2 How the Prompt Is Built

The Gemini prompt is constructed dynamically from the user's actual trained results. It injects the real R^2, RMSE, MAE, SHAP feature rankings, ANN overparameterisation diagnostics, and GPR calibration coverage into the prompt before each call. The model never receives hardcoded BMS1 or BMS2 values — every sentence in the generated report is grounded in the user's uploaded data.

The prompt also injects the BMS1 reference values for comparison, enabling the AI to automatically contextualise the user's results against the published benchmark (e.g., 'Your XGBoost achieved R^2=0.83, matching the BMS2 reference and exceeding the BMS1 baseline by 12%').

10.3 Regulatory Alignment

The Scientific report type includes explicit ICH Q8 (Pharmaceutical Development) and ICH Q9 (Quality Risk Management) references. The prompt instructs Gemini to:

- Use the Quality-by-Design (QbD) framing from ICH Q8 when discussing the relationship between bioprocess parameters and fucosylation.
- Reference the Design Space concept when discussing GPR uncertainty intervals.
- Cite the relevant literature (Raju et al. 2000, Hossler et al. 2009, Lundberg & Lee 2017) inline in the text.
- End with a concrete deployment recommendation aligned with FDA AI/ML guidance.

The result is a report that a process development scientist could submit as a working document in an IND or BLA filing, with appropriate caveats about the synthetic nature of the training data.

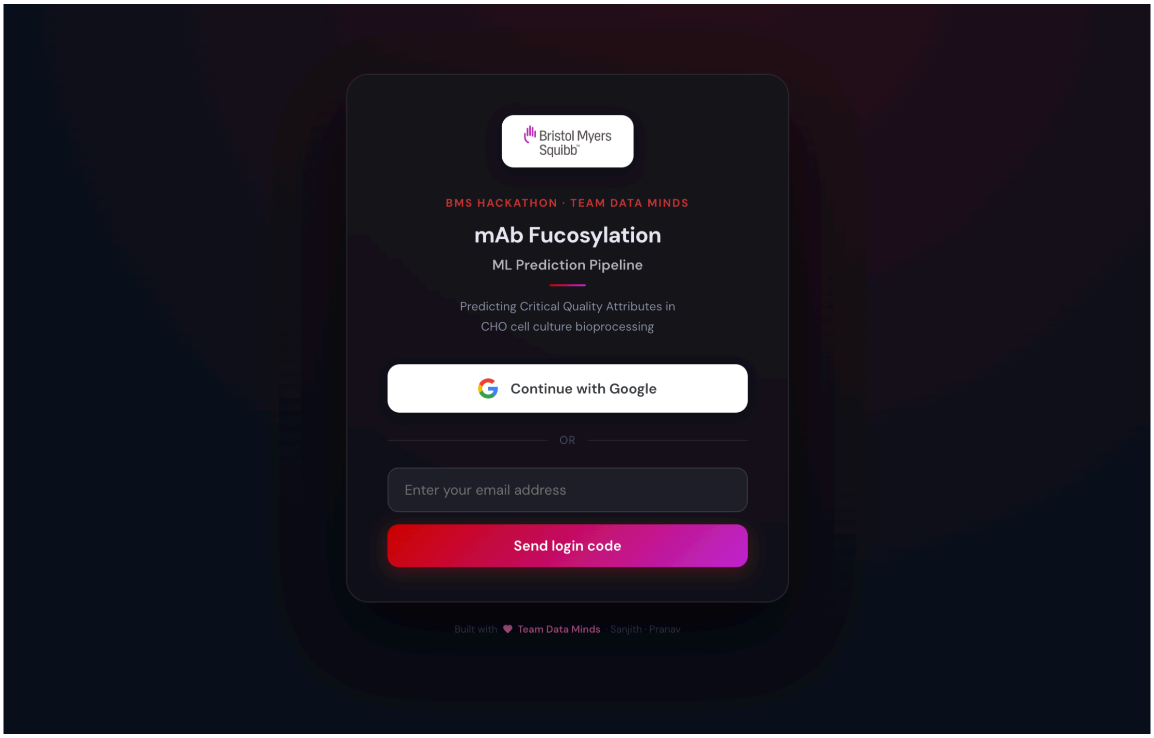
11. Future Extensions

- **Real-Time Monitoring**: Connect the trained model to a LIMS or MES via the REST API. As each process measurement comes in during a batch run, the model predicts end-of-batch fucosylation. Alerts fire when predictions drift outside specification.
- **Multi-CQA Extension**: The same pipeline handles any Critical Quality Attribute — galactosylation, sialylation, HMW species, or product yield — with no code changes.
- **Transfer Learning Across Products**: A model trained on one mAb product can be fine-tuned on a small dataset from a new product, using the learned prior rather than starting from scratch.
- **Uncertainty-Guided Batch Release**: GPR confidence intervals drive risk-based release decisions. Narrow interval = provisional release. Wide interval = flag for additional QC testing.
- **Federated Learning**: In a multi-site setting, models can be trained locally at each site and aggregated without sharing proprietary batch data between sites.

12. Dashboard UI — Feature Walkthrough

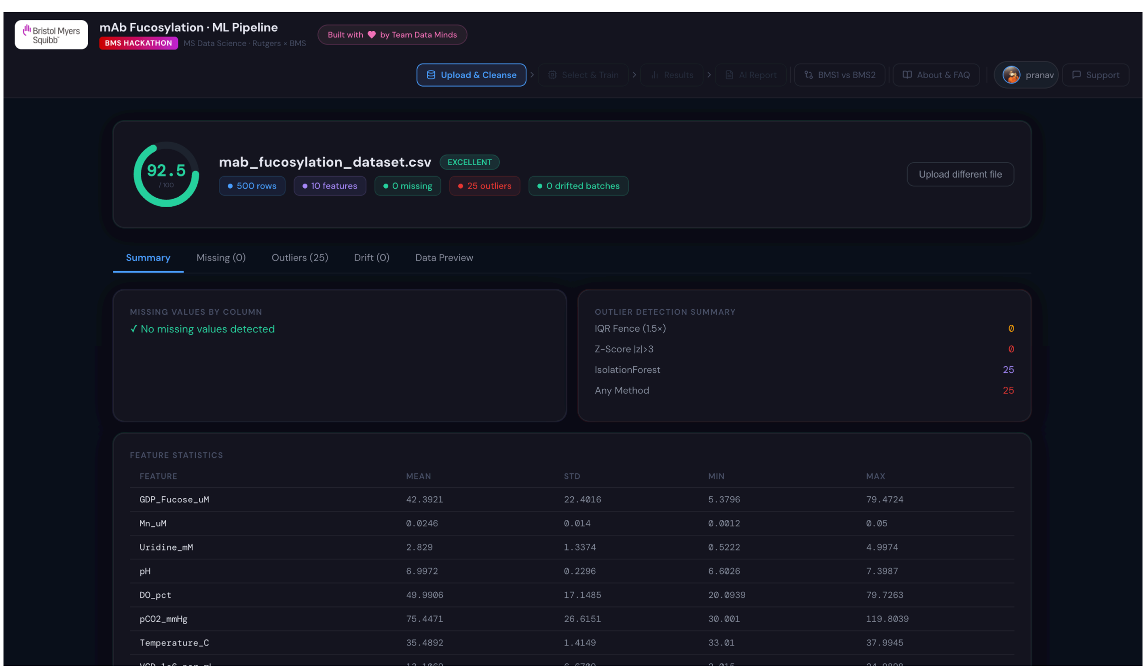
The following placeholders indicate where screenshots of the live dashboard should be inserted. Each screenshot illustrates a specific feature of the platform for a specific scientific or operational purpose. The dashboard is accessible at http://localhost:3000 after following the setup instructions in the project README.

12.1 Login Page



Two passwordless sign-in options. Google OAuth (one-click, no password) for users with Google accounts. Email OTP (6-digit code, 10-minute expiry) for all other email addresses. No credentials are stored by the application.

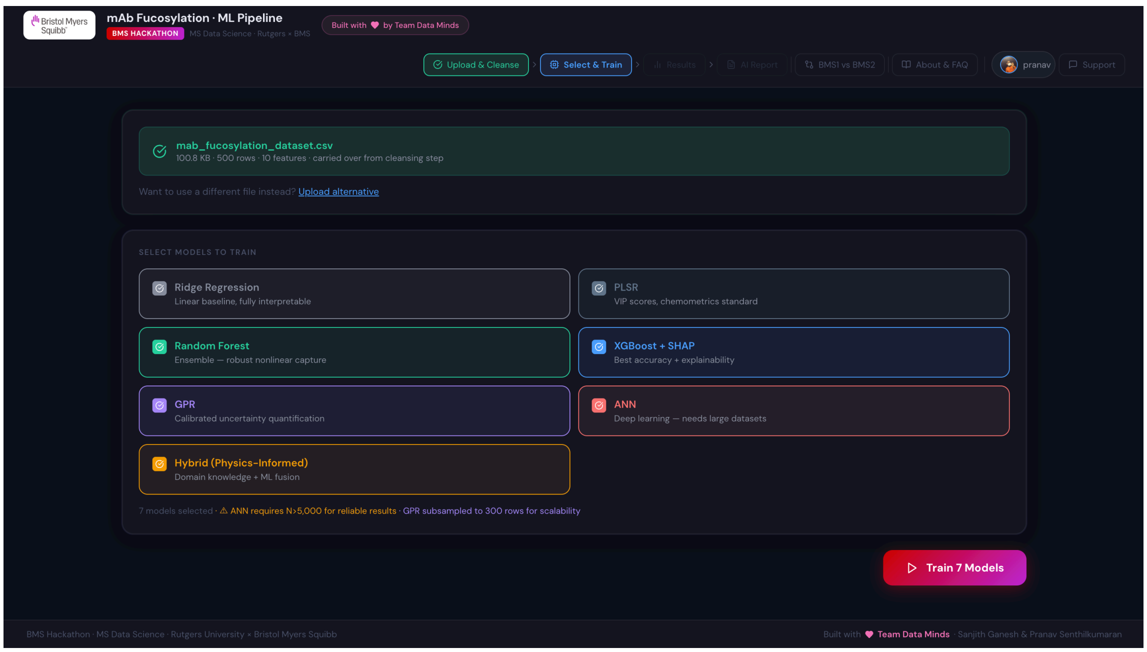
12.2 Upload & Cleanse



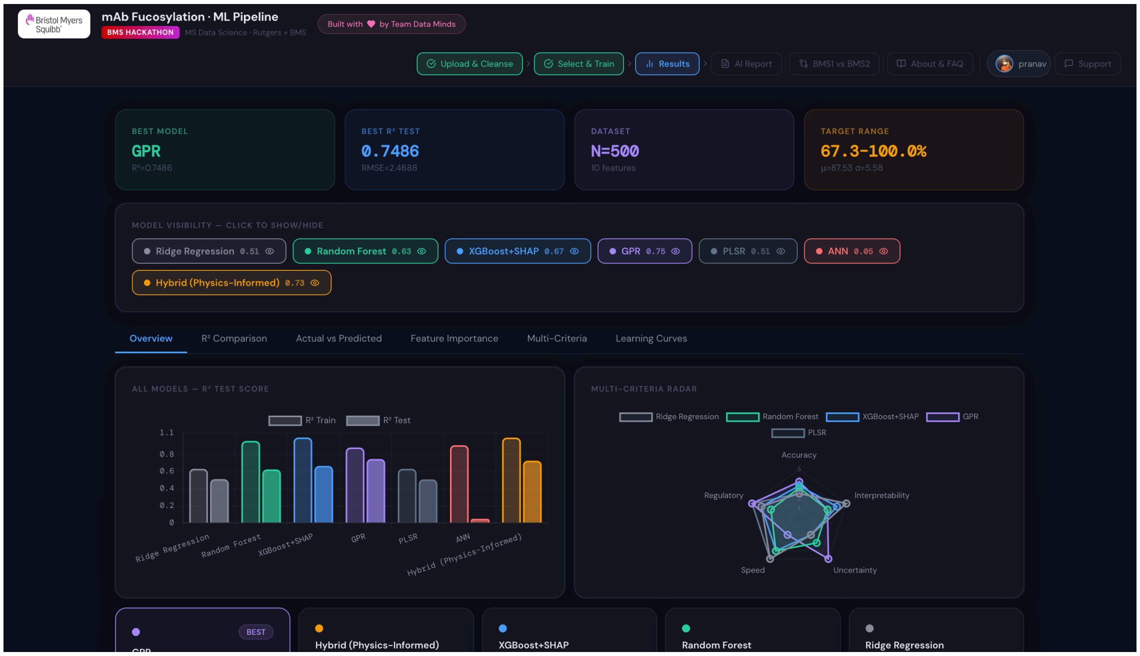
CSV dropzone with instant quality analysis. The Summary tab shows overall score, row count, feature count, and counts for missing values, outliers, and drifted batches. The Missing tab breaks down MCAR vs MNAR patterns per column. The Outliers tab shows IQR, Z-score, and IsolationForest results separately. The Drift tab shows per-batch KS test p-values. The Data Preview shows summary statistics for each feature.

12.3 Model Selection & Training

The user selects any combination of the 7 models. The uploaded file carries over — no re-upload. Clicking Train sends the file and model selection to the backend, which trains each model on the real data and returns results. A progress indicator shows which model is currently training.



12.4 Results Dashboard

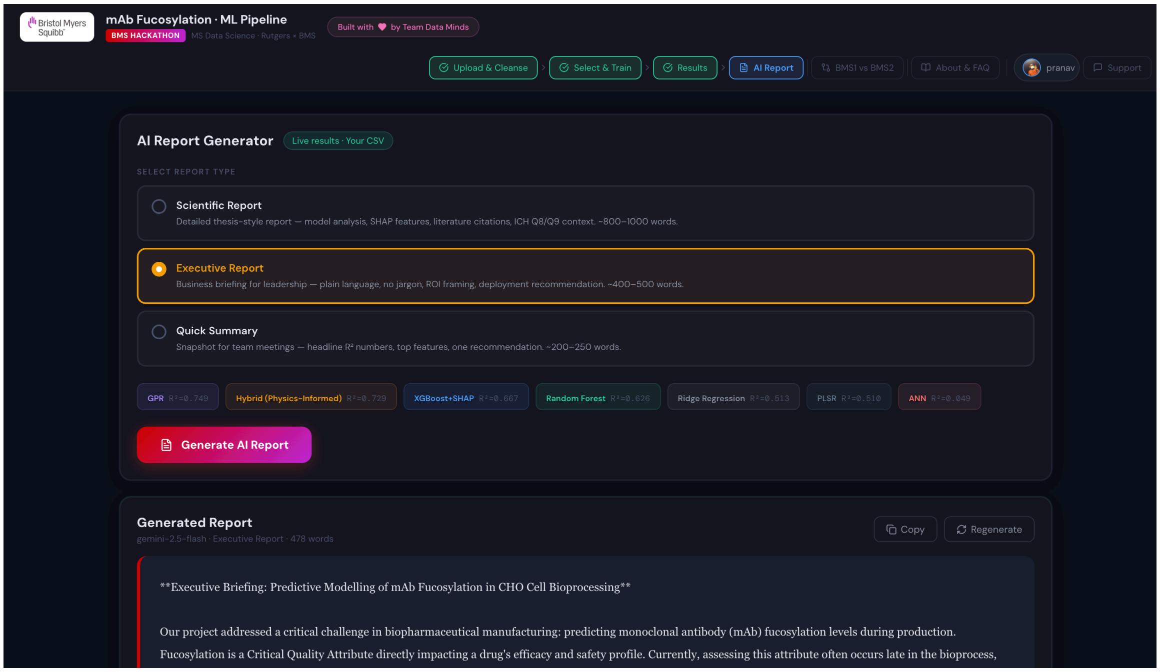


The main results view shows four KPI cards (best model, best R^2, dataset size, target range), model visibility toggles that hide/show individual models across all charts, and a grouped R^2 bar chart comparing train and test performance across all selected models.

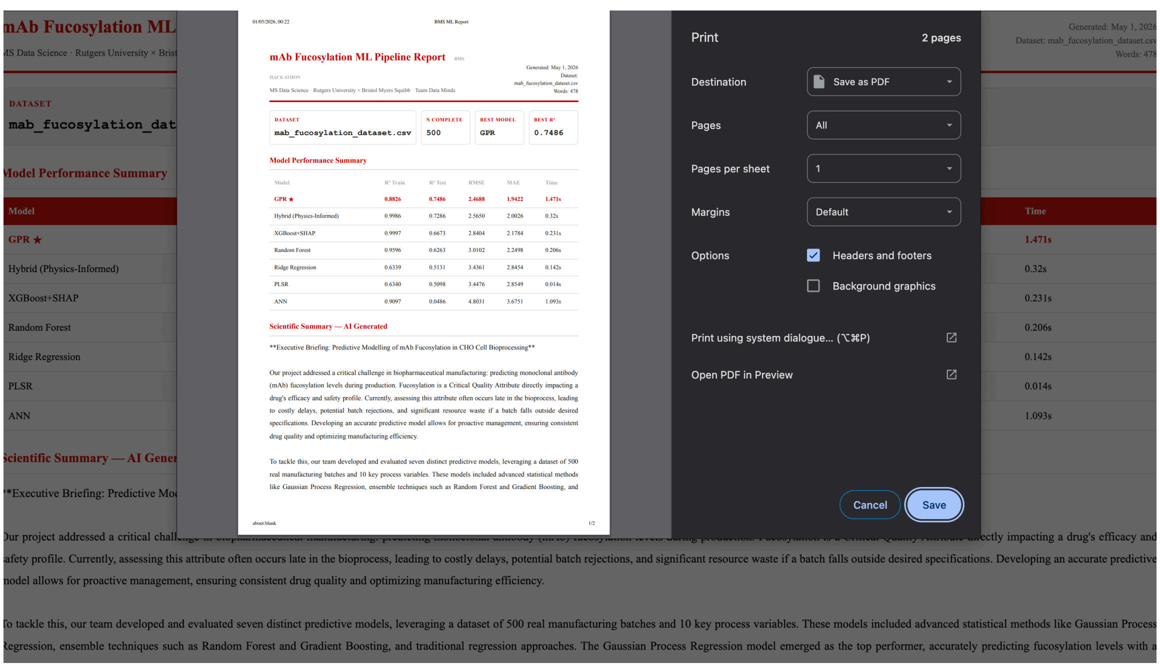
Clicking any model name opens its actual vs predicted scatter plot. Points tight to the diagonal indicate good generalisation. The scatter for the ANN at N=500 would show a horizontal cloud (random predictions); at N=10,000 it recovers to near-diagonal.

Three feature importance methods available depending on model selected. XGBoost shows SHAP mean absolute values per feature. PLSR shows VIP scores (>1.0 = significant). Random Forest shows MDI and permutation importance side by side with Spearman correlation between them.

12.5 AI Report Generator

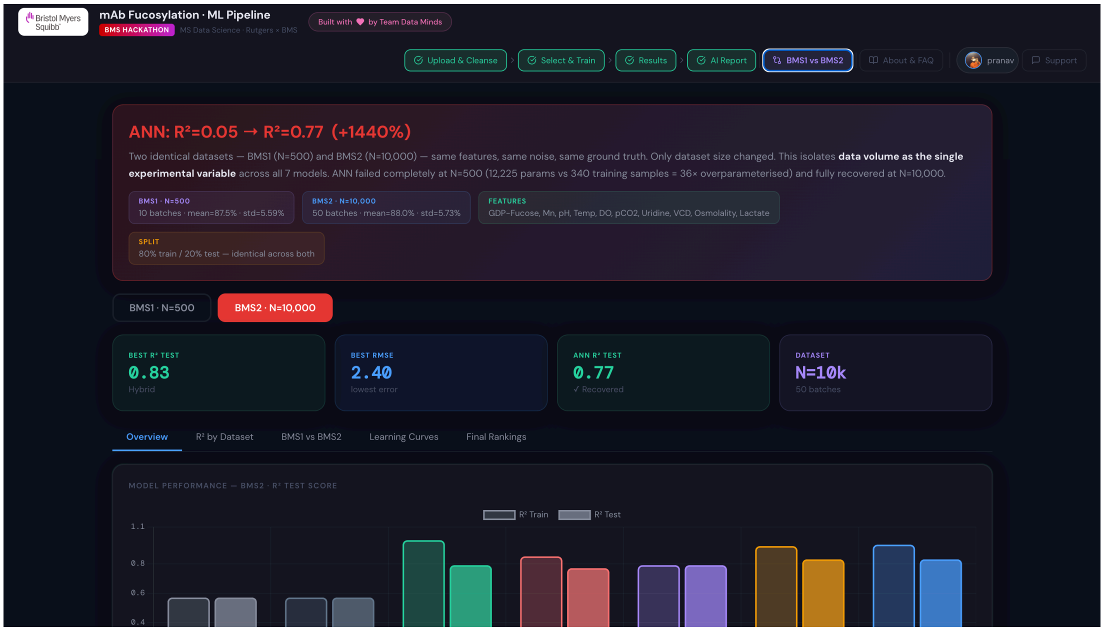


Three report type cards with radio-button selection. Scientific (800–1,000 words), Executive (400–500 words), Quick Summary (200–250 words). All generated from the user's actual trained results. The Generate AI Report button triggers a Gemini 2.5 Flash API call with a prompt dynamically built from the real model metrics.



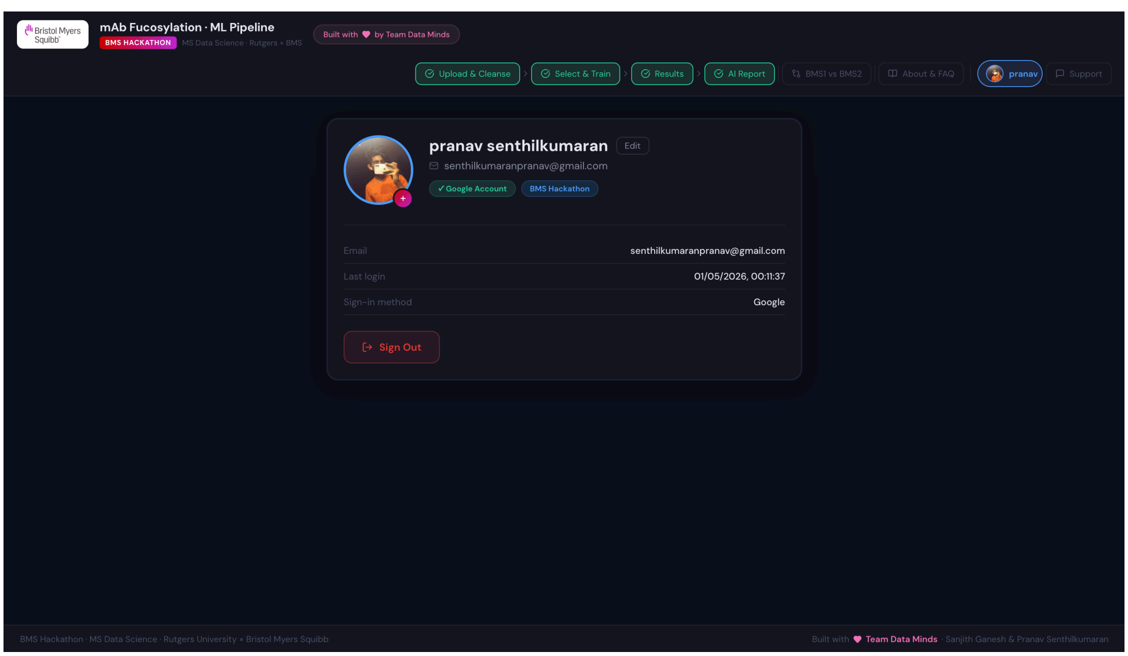
The generated report displays in a styled panel with word count indicator. Copy to clipboard and Export as PDF (browser print to PDF, A4 format) are available.

12.6 BMS1 vs BMS2 Static Reference

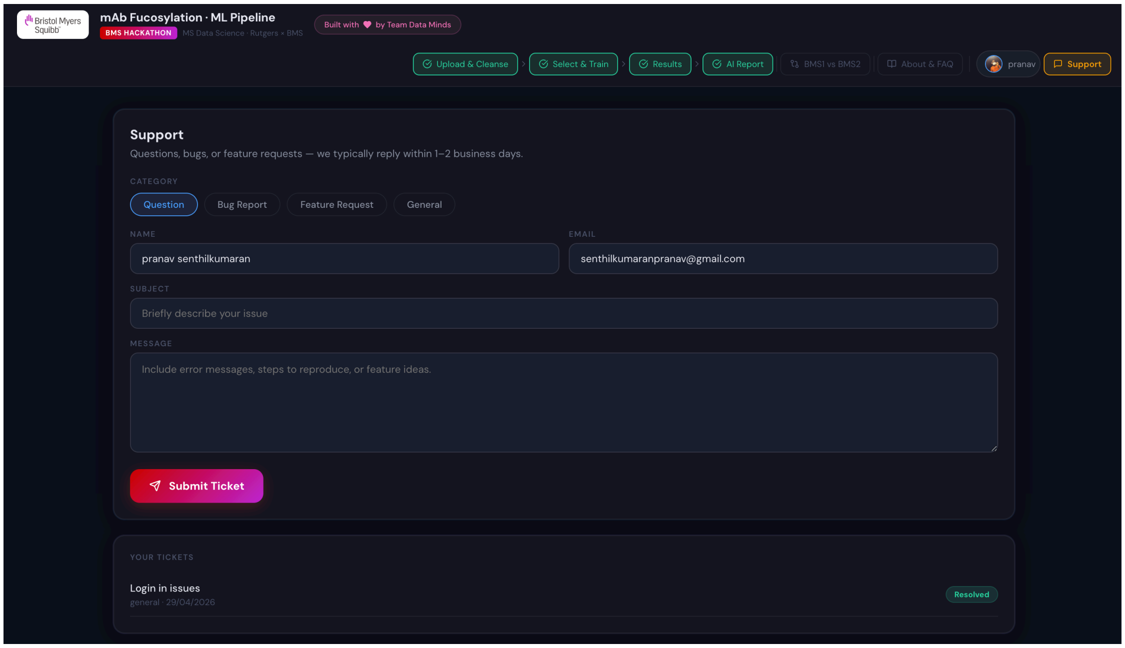


A static reference page always accessible from the navigation bar. Five tabs: Overview (model cards ranked by R^2), R^2 by Dataset (full metrics table), BMS1 vs BMS2 (side-by-side grouped bar chart and delta table), Learning Curves (R^2 vs training dataset size showing ANN's dramatic recovery), and Final Rankings (five-category recommendation table).

12.7 Profile & Support



Profile page shows Google account photo (or OTP user initials), editable display name, email, last login time, and sign-in method. Sign Out clears the localStorage session immediately.



Support form with four category tags (Question, Bug Report, Feature Request, General). Submitted tickets are tracked locally with open/resolved status. Clicking Mark Resolved updates the ticket status. All submissions are delivered via Gmail SMTP to the team inbox.

13. Discussion & Recommendations

13.1 What the bench actually proved

The seven-model bench, run twice across a clean 20× dataset-size jump, produces five conclusions that are not just statements about this dataset but transferable to any CHO bioprocessing modelling problem of this shape:

- **Linear models hit a structural ceiling.** Ridge and PLSR top out at $R^2 \approx 0.57$ regardless of data volume because the underlying physics has quadratic optima and a multiplicative interaction. No amount of data fixes that. The interpretability they offer is real, but they are not predictively competitive.
- **Tree ensembles are the safe default.** Random Forest and XGBoost are competitive at small n ($R^2 \approx 0.62$–0.67) and excellent at large n ($R^2 \approx 0.79$–0.83). They are robust under noise, interpretable through SHAP, and align with FDA AI/ML guidance. If only one model gets deployed, it should be XGBoost+SHAP.
- **GPR is the right tool when data is scarce and uncertainty matters.** On BMS1 the GPR was both the most accurate model and the only one with calibrated prediction intervals. For early-development bioprocesses where $N < 1{,}000$ and ICH Q8 Design Space work is the goal, GPR is the right default — accept the $O(n^3)$ ceiling and use a subsample at scale.

- **ANNs need data, full stop.** The 1,440% R^2 gain from BMS1 to BMS2 is the cleanest empirical statement of this you can ask for. Pharma bioprocess datasets are usually below the threshold where unconstrained ANNs work; deploy them only when the data is genuinely large or use a structured network with explicit priors.
- **Physics-informed features earn their seat.** Five engineered features capture roughly three-quarters of full model performance — that is a dimensionality halving with a 25% accuracy cost, which in regulated settings often comes out ahead because every variable in a model has to be justified to a regulator.

13.2 Recommended deployment strategy

- **Production model.** XGBoost+SHAP, retrained on real BMS data ingested through the Stage 1 cleansing module. Top accuracy and explainability at the data scales BMS production reaches.
- **Early-development / Design Space model.** GPR with RBF kernel on the cleansed data. Use the credible intervals for ICH Q8 Design Space boundary definition.
- **Regulatory baseline.** PLSR. Always trained, always reported, always submitted alongside the chosen model — even when its accuracy is mediocre — because it is the chemometrics gold standard and the easiest model to defend in a filing.

13.3 Limitations and what to validate next

- **Coefficient magnitudes are assumed, not measured.** The mechanistic equation in Section 3.3 is grounded in published biochemistry but the specific coefficients are calibrated, not fit to real BMS data. The model bench is therefore a methodology validator, not a deployment-ready predictor.
- **Inputs are independent in the synthetic data.** Real bioprocess inputs are correlated — pH and pCO_2 track each other, VCD and lactate are linked. The next step is to replay the bench on a dataset with realistic input correlations, or to validate against a published CHO dataset where such correlations are present.
- **Batch-to-batch variability is modelled as a random intercept only.** Real batches also show correlated drift in slopes (e.g., the temperature effect itself can shift batch-to-batch). A mixed-effects extension would be the natural next iteration.
- **Time-series modelling is out of scope for Stage 2.** The time-series CSV is generated and saved but not fed through the seven-model bench, all of which are steady-state. A dedicated time-series module (LSTM, state-space, or Gaussian-Process latent variable) would be a natural Stage 4.

14. References

[1] Raju TS, Briggs JB, Borge SM, Jones AJS. (2000). *Species-specific variation in glycosylation of IgG: evidence for the species-specific sialylation and branch-specific galactosylation and importance for engineering recombinant glycoprotein therapeutics.* Glycobiology 10(5): 477–486. https://doi.org/10.1093/glycob/10.5.477

[2] Hossler P, Khattak SF, Li ZJ. (2009). *Optimal and consistent protein glycosylation in mammalian cell culture.* Glycobiology 19(9): 936–949. https://doi.org/10.1093/glycob/cwp079

[3] Jedrzejewski PM, del Val IJ, Constantinou A, Dell A, Haslam SM, Polizzi KM, Kontoravdi C. (2014). *Towards controlling the glycoform: a model framework linking extracellular metabolites to antibody glycosylation.* International Journal of Molecular Sciences 15(3): 4492–4522. https://doi.org/10.3390/ijms15034492

[4] Sha S, Agarabi C, Brorson K, Lee D-Y, Yoon S. (2016). *N-Glycosylation Design and Control of Therapeutic Monoclonal Antibodies.* Trends in Biotechnology 34(10): 835–846. https://doi.org/10.1016/j.tibtech.2016.02.013

[5] Von Stosch M, Davy S, Francois K, Galvanauskas V, Hamelink JM, Luebbert A, Mayer M, Oliveira R, O'Kennedy R, Rice P, Glassey J. (2014). *Hybrid modeling for quality by design and PAT — benefits and challenges of integration of data-driven and mechanistic bioprocess models.* Biotechnology Journal 9(6): 719–726. https://doi.org/10.1002/biot.201300385

[6] Lundberg SM, Lee S-I. (2017). *A Unified Approach to Interpreting Model Predictions.* Advances in Neural Information Processing Systems (NeurIPS) 30. https://arxiv.org/abs/1705.07874

[7] Strobl C, Boulesteix A-L, Zeileis A, Hothorn T. (2007). *Bias in random forest variable importance measures: illustrations, sources and a solution.* BMC Bioinformatics 8: 25. https://doi.org/10.1186/1471-2105-8-25

[8] ICH Harmonised Tripartite Guideline Q8(R2). (2009). *Pharmaceutical Development.* International Council for Harmonisation. https://www.ich.org/page/quality-guidelines

[9] ICH Harmonised Tripartite Guideline Q9. (2005). *Quality Risk Management.* International Council for Harmonisation. https://www.ich.org/page/quality-guidelines

[10] FDA. (2023). *Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence/Machine Learning (AI/ML)-Enabled Device Software Functions.* U.S. Food and Drug Administration. https://www.fda.gov/regulatory-information/search-fda-guidance-documents

[11] Ganesh S, Senthilkumaran P. (2026). *Predictive Modelling of mAb Fucosylation in CHO Cell Bioprocessing — GitHub repository.* M.S. Data Science, Rutgers University. https://github.com/SanjithGanesh/Predictive-Modelling-of-mAb-Fucosylation---BMS

15. Appendix — Dataset Schema, Hyperparameters, File Index

A.1 Dataset files generated by 00_dataset_generator.py

File	Rows × Cols (BMS1)	Rows × Cols (BMS2)	Description
mab_fucosylation_dataset.csv	500 × 12	10,000 × 12	Primary dataset — used for model training
mab_fucosylation_MCAR.csv	500 × 12	10,000 × 12	5% values missing completely at random
mab_fucosylation_MNAR.csv	500 × 12	10,000 × 12	Sensor-failure missingness on DO and pCO_2
mab_fucosylation_noise_low.csv	500 × 12	10,000 × 12	0.5× extra measurement noise
mab_fucosylation_noise_medium.csv	500 × 12	10,000 × 12	1.0× extra measurement noise
mab_fucosylation_noise_high.csv	500 × 12	10,000 × 12	2.0× extra measurement noise
mab_fucosylation_noise_extreme.csv	500 × 12	10,000 × 12	4.0× extra measurement noise
mab_fucosylation_timeseries.csv	280 × 13	1,400 × 13	20 / 100 batches × 14 days
ground_truth_params.csv	1 × 14	1 × 14	Ground-truth coefficients for validation

Table 30: *All dataset files generated by Script 00 in each folder.*

A.2 Hyperparameter summary

Model	Key hyperparameters
Ridge	RidgeCV, $\alpha \in$ logspace(−4, 4, 50), 5-fold CV
PLSR	components $\in$ [1..10], optimal selected by 5-fold CV
Random Forest	n_estimators=300, max_depth=15, min_samples_leaf=5, OOB on
ANN	hidden=(128,64,32), ReLU, Adam, α=1e-4, max_iter=2000, early stop, patience 50
GPR	RBF + WhiteKernel, n_restarts=3, α=0.1, training subsample = 300
Hybrid	Random Forest backbone with 5 engineered features + 10 raw features
XGBoost	n_estimators=300, max_depth=5, lr=0.1, subsample=0.8

Table 31: *Hyperparameter configuration for each model. Identical across BMS1 and BMS2.*